

RECIPE	example	fact	definition	trap	table
The task says...					
“Berechnen Sie den Grenzwert” (0/0)		\$5.3 Taylor-plug, beats l’Hospital			
“Grenzwert einer Folge”		\$3.2, or \$3.3 if recursive			
“Konvergiert die Reihe?”		\$4.2 decision tree			
“Konvergenzradius / welche x”		\$5.1 radius recipe			
“Bestimmen Sie $\int f\,dx$ ”		\$9.1 which method			
“allgemeine Lösung der DGL”		\$10.1 which method			
“Kurvendiskussion / Extrema”		\$8.4 full recipe			

T = the statement is **true**, F = it is **false**; the statement itself is coloured to match.

The strip along the **top edge** lists every section with its page; the **dark** entries and the **edge tabs** are the ones live on this page.

1 Foundations: sup/inf, induction, estimates, vectors

1.1 Quantifiers, negation, and the standard vocabulary

DEF: the vocabulary every exercise uses — defined here, used everywhere

- Intervals: *open* (a, b) (no endpoints), *closed* $[a, b]$ (both), half-open $[a, b)$. $|I|$ = length.
- Bounded: $\exists C : |x| \leq C$ for all x in the set. Bounded above / below separately.
- Compact (*kompakt*) in $\mathbb{R} \iff$ **closed and bounded** (Heine–Borel). This one word is the hypothesis of IVT, Weierstrass and Heine (§6.4); drop it and all three fail.
- Neighbourhood / ε -ball of x : the interval $(x - \varepsilon, x + \varepsilon)$. interior point: some ε -ball fits entirely inside the set; **boundary point**: every ε -ball meets the set *and* its complement.
- Null sequence: $a_n \rightarrow 0$. Subsequence (a_{n_k}) : indices **strictly increasing**, $n_1 < n_2 < \dots$ (so $n_k \geq k$).
- Dense in $\mathbb{R}$: every interval, however short, contains a point of the set. $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ both are.

► RECIPE: negate any definition mechanically

Rule: swap $\forall \leftrightarrow \exists$ left to right, **keep the order**, negate the final inequality ($<$ becomes $\geq$). !The order is the whole content: $\forall x \exists \delta$ (continuity, δ may depend on x) is a *different theorem* from $\exists \delta \forall x$ (uniform, **one** δ for all x).

- $a_n \not\rightarrow a$: $\exists \varepsilon > 0 \forall N \exists n \geq N : |a_n - a| \geq \varepsilon$.
- not Cauchy**: $\exists \varepsilon > 0 \forall N \exists m, n \geq N : |a_n - a_m| \geq \varepsilon$.
- f **not continuous** at x_0 : $\exists \varepsilon > 0 \forall \delta > 0 \exists x : |x - x_0| < \delta$ and $|f(x) - f(x_0)| \geq \varepsilon$.
- f **not uniformly continuous**: $\exists \varepsilon > 0 \forall \delta > 0 \exists x, y : |x - y| < \delta$ and $|f(x) - f(y)| \geq \varepsilon$.
- $f_n \not\rightarrow f$: $\exists \varepsilon > 0 \forall N \exists n \geq N \exists x : |f_n(x) - f(x)| \geq \varepsilon$.

In practice the last two become: build x_n, y_n resp. x_n and show the gap does not vanish.

1.2 sup / inf / max / min (Supremum, Infimum)

DEF: the four words

$M \subseteq \mathbb{R}$ nonempty. **upper bound** s : $x \leq s \forall x \in M$. $\sup M$ = least upper bound. $\max M = \sup M$ only if $\sup M \in M$. Mirrored for inf/min.
Completeness axiom: every nonempty M bounded above has $\sup M \in \mathbb{R}$. This is the one property $\mathbb{Q}$ lacks and the engine behind IVT, Bolzano–Weierstrass and monotone convergence.

FACT: ε -characterisation (use it in every sup proof)

$s = \sup M \iff$ (i) $x \leq s \forall x \in M$, and (ii) $\forall \varepsilon > 0 \exists x \in M : x > s - \varepsilon$.
For a function: $\sup_{x \in D} f(x)$ is the sup of the image set $f(D)$; it is a max iff attained.

► RECIPE: find sup/inf and prove it

- Guess s (the largest/smallest value the formula can approach).
- Bound: show $x \leq s$ for all $x \in M$ by algebra.
- Least: given $\varepsilon > 0$ produce $x \in M$ with $x > s - \varepsilon$ (usually choose $n > 1/\varepsilon$).
- Attained? $s \in M \Rightarrow \max = s$; otherwise max does not exist.

► RECIPE: a set is given by a formula: open? closed? bounded? sup? inf?

Open: every point has an ε -ball still inside M . Closed: the complement is open $\iff$ every convergent sequence in M has its limit in M .

- Given by an inequality on a continuous g : $\{g > c\}$ and $\{g < c\}$ are **open**; $\{g \geq c\}$, $\{g \leq c\}$, $\{g = c\}$ are **closed**. Solve the inequality into intervals and read it off.
- Given by a sequence formula $\{a_n : n \in \mathbb{N}\}$: check monotonicity. Monotone decreasing $\Rightarrow \sup = \max = a_1$ (attained), $\inf = \lim a_n$ (**not** attained). Increasing $\Rightarrow$ mirrored. Not monotone $\Rightarrow$ split by parity first (§3.4).
- Bounded? both sup and inf finite. Unbounded above $\Rightarrow$ write $\sup = +\infty$.

!“*neither open nor closed*” is the usual trap answer: $[a, b)$ qualifies, but $\mathbb{R}$ and $\emptyset$ are **both** open and closed, and a finite set is closed.

Ex. $A = \{x : |x + 1|^3 > 2\} = \{[x + 1]^3 > \sqrt[3]{2}\} = (-\infty, -1 - \sqrt[3]{2}) \cup (-1 + \sqrt[3]{2}, \infty)$: **open**, unbounded, $\sup A = +\infty$, $\inf A = -\infty$.

Ex. $B = \{\frac{1}{n+1} - \frac{1}{n+2}\} = \{\frac{1}{(n+1)(n+2)}\}$: decreasing $\Rightarrow$ max at the first index, $\inf = 0$ not attained.

FACT: Archimedes and density

Archimedean: $\forall x \in \mathbb{R} \exists n \in \mathbb{N} : n > x$; equivalently $\frac{1}{n} \rightarrow 0$. $\mathbb{N}$ is unbounded in $\mathbb{R}$.
 $\mathbb{Q}$ **dense**: between any $a < b$ there is a rational *and* an irrational. Used to build the Dirichlet counterexamples.

TRAP: sup/inf in multiple choice

- sup exists as a *real number* only for a bounded-above set. Unbounded $\Rightarrow$ write $\sup = +\infty$, i.e. no real sup.
- min M need **not** be an accumulation point of the sequence: $x_0 = 0$, $x_n = 1 + \frac{1}{n}$ attains its min once.
- $\inf M = \sup M \Rightarrow M$ is a single point $\Rightarrow$ the sequence is constant $\Rightarrow$ converges. T
- monotone *and* bounded $\Rightarrow$ converges to $\sup$ (incr.) / $\inf$ (decr.). Monotone alone may run to $\pm\infty$.

1.3 Absolute value and inequalities

► RECIPE: solve $| \cdot |$ inequalities

$|x - a| < r \iff x \in (a - r, a + r); \quad |x - a| > r \iff x < a - r \text{ or } x > a + r.$
 $|f(x)| < g(x) \iff -g(x) < f(x) < g(x)$ (and $g > 0$).
General: split at every zero of the inside, solve on each piece, intersect with that piece.

FACT: the inequalities you actually use

- Triangle**: $|a + b| \leq |a| + |b|$; **reverse**: $||a| - |b|| \leq |a - b|$.
- Bernoulli**: $(1 + x)^n \geq 1 + nx$ for $x \geq -1$.
- AM–GM: $\sqrt[n]{ab} \leq \frac{a+b}{2}$ ($a, b \geq 0$).
- Young: $ab \leq \frac{a^2}{2} + \frac{b^2}{2}$, sharper $ab \leq \frac{a^2}{2} + \frac{b^2}{2c}$.
- Cauchy–Schwarz**: $|\langle x, y \rangle| \leq \|x\| \|y\|$.

Always available: $|\sin t| \leq 1$, $|\sin t| \leq |t|$ (strict for $t \neq 0$), $|\cos t| \leq 1$, $e^t \geq 1 + t$, $\ln t \leq t - 1$.

► RECIPE: estimate without panic

Target is always $|\text{expr}| \leq$ (something $\rightarrow 0$).

- Numerator: make it *bigger* (drop negative terms, bound $\sin/\cos$ by 1).
- Denominator: make it *smaller* (drop positive terms). Never the reverse.
- Roots: conjugate, $\sqrt{A} - \sqrt{B} = \frac{A-B}{\sqrt{A}+\sqrt{B}}$.
- Sums: bound every term by the largest, then multiply by the count.

FACT: small- x behaviour: the six approximations, and the inequalities that prove them

For $x \rightarrow 0$ (*and only then*), each function may be replaced by its first Taylor term; the error is one order higher (full list §5.4, §13.3).

$\sin x \approx x$	$x - \frac{x^3}{6} + O(x^5)$	exact : $ \sin x \leq x $ always
$\tan x \approx x$	$x + \frac{x^3}{3} + O(x^5)$	$\tan x \geq x$ on $[0, \frac{\pi}{2})$
$1 - \cos x \approx \frac{x^2}{2}$	$\frac{x^2}{2} - \frac{x^4}{24} + O(x^6)$	$0 \leq 1 - \cos x \leq \frac{x^2}{2}$
$e^x \approx 1 + x$	$1 + x + \frac{x^2}{2} + O(x^3)$	exact : $e^x \geq 1 + x$ for all x
$\ln(1 + x) \approx x$	$x - \frac{x^2}{2} + O(x^3)$	$\frac{x}{1+x} \leq \ln(1 + x) \leq x$
$(1 + x)^a \approx 1 + ax$	$1 + ax + O(x^2)$	Bernoulli : $\geq$ for $a \geq 1$

!Use the exact column when you must *prove* an estimate; the $\approx$ column only inside a limit. Substituting $x = \frac{1}{n}$ turns every row into a sequence estimate, e.g. $n \sin \frac{1}{n} \rightarrow 1$.

1.4 Induction (vollständige Induktion)

► RECIPE: proof by induction

- Base: verify $n = n_0$ explicitly.
- Hypothesis: assume $A(n)$ for one fixed $n \geq n_0$.
- Step: write the $n+1$ statement, isolate the n part, substitute the hypothesis, then estimate to the target. Say where you used it.

Helpers: $\sum_{k=1}^n k = \frac{n(n+1)}{2}$, $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$, $\sum_{k=0}^n q^k = \frac{1-q^{n+1}}{1-q}$, $(1 + x)^n \geq 1 + nx$.

1.5 Vectors, inner product, norms in $\mathbb{R}^n$

DEF: inner product, norm, distance

$\langle x, y \rangle = x \cdot y = \sum_{i=1}^n x_i y_i; \quad \|x\| = \sqrt{\langle x, x \rangle} = \sqrt{\sum x_i^2}; \quad d(x, y) = \|x - y\| = \sqrt{\sum (x_i - y_i)^2}.$

FACT: Cauchy–Schwarz and the norm axioms

$|\langle x, y \rangle| \leq \|x\| \|y\|$, with **equality iff** x, y are **linearly dependent**. This is exactly what makes the triangle inequality true.
A **norm** satisfies: $\|x\| \geq 0$ and $\|x\| = 0 \iff x = 0$; $\|cx\| = |c| \|x\|$; $\|x + y\| \leq \|x\| + \|y\|$. Hence $\|x - z\| \leq \|x - y\| + \|y - z\|$ for all x, y, z .

2 Complex numbers and polynomials

2.1 Forms, conversion, arithmetic

DEF: the two forms

$z = a + ib = re^{i\varphi} = r(\cos \varphi + i \sin \varphi); \quad \bar{z} = a - ib, \quad |z| = r = \sqrt{a^2 + b^2}, \quad z\bar{z} = |z|^2.$
 $Re\, z = \frac{z+\bar{z}}{2}, \quad Im\, z = \frac{z-\bar{z}}{2i}$. Euler: $e^{i\varphi} = \cos \varphi + i \sin \varphi$, $e^{i\pi} = -1$, $e^{2\pi i} = 1$.

► RECIPE: convert both ways — the quadrant table settles it

Polar $\rightarrow$ Cartesian (never ambiguous): $a = r \cos \varphi$, $b = r \sin \varphi$.
Cartesian $\rightarrow$ polar: $r = \sqrt{a^2 + b^2}$, and read φ off the sign of a, b — **do not think, look it up**:

$a > 0$ (right half)	$\varphi = \arctan \frac{b}{a}$	quadrant I / IV
$a < 0, b \geq 0$	$\varphi = \arctan \frac{b}{a} + \pi$	quadrant II
$a < 0, b < 0$	$\varphi = \arctan \frac{b}{a} - \pi$	quadrant III
$a = 0$	$\varphi = +\frac{\pi}{2}$ if $b > 0$, $-\frac{\pi}{2}$ if $b < 0$	φ undefined at $z = 0$

Why: arctan only ever returns a value in $(-\frac{\pi}{2}, \frac{\pi}{2})$, i.e. it always answers in the *right* half-plane; for $a < 0$ you must rotate by π to reach the correct half. The **sign** of that π is chosen so the answer stays in the principal range $(-\pi, \pi]$ — $+\pi$ would overshoot past π when $b < 0$. !A φ outside $(-\pi, \pi]$ is not wrong, just not principal: add or subtract 2π to bring it back.
Ex. $z = -1 - i$: $r = \sqrt{2}$, $\arctan \frac{-1}{-1} = \frac{\pi}{4}$, third quadrant $\Rightarrow \varphi = \frac{\pi}{4} - \pi = -\frac{3\pi}{4}$. ($+\pi$ would give $\frac{5\pi}{4} > \pi$.)

FACT: arithmetic rules

Multiply/divide in polar: $r_1 e^{i\varphi_1} \cdot r_2 e^{i\varphi_2} = r_1 r_2 e^{i(\varphi_1 + \varphi_2)}$, $\frac{z}{w} = \frac{r_z}{r_w} e^{i(\varphi_z - \varphi_w)}$.
Divide in Cartesian: multiply by $\frac{\bar{w}}{w}$: $\frac{z}{w} = \frac{z\bar{w}}{|w|^2}$.
Conjugate: $\bar{z + w} = \bar{z} + \bar{w}$, $\overline{z\bar{w}} = z w$, $\overline{z/w} = \bar{z}/\bar{w}$, $\bar{\bar{z}} = z$, $z \in \mathbb{R} \iff z = \bar{z}$, $|\bar{z}| = |z|$.
de Moivre: $(\cos \varphi + i \sin \varphi)^n = \cos(n\varphi) + i \sin(n\varphi)$.

► RECIPE: use Euler to derive trig identities on the spot

$\cos x = \frac{e^{ix} + e^{-ix}}{2}, \sin x = \frac{e^{ix} - e^{-ix}}{2i}$. Multiply the exponentials, collect.
Ex. $\cos^2 x = \left(\frac{e^{ix} + e^{-ix}}{2}\right)^2 = \frac{e^{2ix} + 2 + e^{-2ix}}{4} = \frac{1 + \cos 2x}{2}$. Same trick gives every double/half-angle formula, so you never have to memorise them.

2.2 Roots and equations

► RECIPE: solve $z^n = w$ (n -th roots)

- Write $w = \rho e^{i\psi}$.
- $z_k = \rho^{1/n} e^{i(\psi + 2\pi k)/n}$ for $k = 0, 1, \dots, n - 1$.
- Exactly n roots, all on the circle of radius $\rho^{1/n}$, spaced by $\frac{2\pi}{n}$.

Roots of unity: $z^n = 1 \Rightarrow z_k = e^{2\pi i k/n}$; their sum is 0 for $n \geq 2$ and their product is $(-1)^{n+1}$.

► RECIPE: quadratic with complex data

$az^2 + bz + c = 0 \Rightarrow z = \frac{-b \pm \sqrt{D}}{2a}, \quad D = b^2 - 4ac.$

- Real coefficients, $D < 0$: $z = -\frac{b}{2a} \pm i \frac{\sqrt{|D|}}{2a}$ (a conjugate pair).
- Complex D : get $\sqrt{D}$ with the n -th-root recipe at $n = 2$ (two roots, take either).

FACT: complex log and general powers + DEF: what a *branch* is

The problem: $e^{i\varphi} = e^{i(\varphi+2\pi k)}$, so $\arg z$ is only fixed **mod** 2π and “log” has *infinitely many* values $\log z = \ln|z| + i(\arg z + 2\pi k)$, $k \in \mathbb{Z}$. That is not a function.
A **branch** = one consistent choice of k . The **principal branch** is the choice $\text{Arg } z \in (-\pi, \pi]$ (the same range as the quadrant table above), giving the genuine function

$$\text{Log } z = \ln|z| + i \text{Arg } z, \quad z^w := e^{w \text{Log } z}.$$

Ex. $\text{Log}(-1) = \ln 1 + i\pi = i\pi$. $i^i = e^{i \text{Log } i} = e^{i \cdot i\pi/2} = e^{-\pi/2} \text{ — real.}$
! Only Log is a function; $\sqrt{zw} = \sqrt{z}\sqrt{w}$ and $\log(zw) = \log z + \log w$ hold for the multivalued log but can be off by $2\pi i$ once you pin down the principal branch. Counterexample $z = w = -1$: $\sqrt{zw} = 1$ but $\sqrt{z}\sqrt{w} = i \cdot i = -1$.

2.3 Polynomials: what is guaranteed

FACT: Fundamental theorem of algebra

p of degree $n \geq 1$ with any complex coefficients has **exactly n roots** in $\mathbb{C}$ **counted with multiplicity**, and factors as $p(z) = a_n \prod_j (z - z_j)^{m_j}$ with $\sum m_j = n$.

TRAP: the three classic multiple-choice errors

- “ n roots, all *distinct*” **F** — only with multiplicity (z^5 has one root of multiplicity 5).
- “complex roots come in conjugate pairs” in *general* **F** — **T only if all coefficients are real**. Degree 5 with complex coefficients need not have any real root.
- “odd degree $\Rightarrow$ a real root” **T** — **only for real coefficients** (IVT on p , which changes sign at $\pm\infty$).

► RECIPE: factor a polynomial over $\mathbb{R}$ / over $\mathbb{C}$

1. Guess a root among the divisors of the constant term (over $\mathbb{Z}$), or spot $z = \pm 1$.
2. Polynomial division by $(z - z_0)$, repeat on the quotient.
3. Over $\mathbb{C}$: keep going until all factors are linear.
4. Over $\mathbb{R}$: pair each non-real root with its conjugate to get an irreducible quadratic $z^2 - 2\text{Re}(z_0)z + |z_0|^2$. Real factorisation = linear factors $\times$ irreducible quadratics only.

3 Sequences (a_n)

3.1 Definitions you must be able to write

DEF: convergence, divergence, boundedness

$a_n \rightarrow a \iff \forall \epsilon > 0 \exists N \in \mathbb{N} \forall n \geq N: |a_n - a| < \epsilon.$
 $a_n \rightarrow \infty \iff \forall K > 0 \exists N \forall n \geq N: a_n > K$ (divergence to ∞ , not convergence).
bounded: $\exists C: |a_n| \leq C \forall n$. **monotone inc.:** $a_{n+1} \geq a_n \forall n$.
explicit $a_n = f(n)$ vs **recursive** $a_{n+1} = f(a_n)$ — the treatment is completely different, see §3.2 vs §3.3.

FACT: the implication chain (know which arrows reverse)

convergent $\Rightarrow$ Cauchy $\Rightarrow$ bounded $\Rightarrow$ has a convergent subsequence (Bolzano–Weierstrass). Only reversal that holds: **Cauchy $\Leftrightarrow$ convergent** in $\mathbb{R}$ (completeness). Also monotone + bounded $\Leftrightarrow$ convergent for *monotone sequences*.

3.2 Limit of an explicitly given a_n

► RECIPE: compute $\lim a_n$: match the shape

1. **Rational in n :** divide top and bottom by the highest power in the *denominator*. deg top < bottom $\Rightarrow 0$; $= \Rightarrow$ ratio of leading coefficients; $> \Rightarrow \pm\infty$.
2. **Roots, $\infty - \infty$:** **multiply by the conjugate** — multiply and divide by $\sqrt{A} + \sqrt{B}$ so that $\sqrt{A} - \sqrt{B} = \frac{A-B}{\sqrt{A}+\sqrt{B}}$; the roots cancel upstairs and the answer falls out. Worked out in full in the EXAMPLE box below, line 2.
3. **n -th roots, n in the exponent:** write $a_n = e^{\ln a_n}$ and take the limit in the exponent. EXAMPLE box below, lines 3–4.
4. $(1 + \frac{\epsilon}{n})^{dn} \rightarrow e^{cd}$.
5. **Factorials, n^n :** use the growth order $\ln n \ll n^a \ll b^n \ll n!$ ($\alpha > 0, b > 1$): the *smaller* one over the bigger one $\rightarrow 0$. EXAMPLE box below, line 5.
6. **log or root of a sum:** factor out the dominant term first. $\log(n + e^{n^2}) = \log(e^{n^2}(1 + ne^{-n^2})) = n^2 + \log(1 + ne^{-n^2}) \sim n^2$.
7. $(-1)^n$ or sin/cos times a null sequence: squeeze $-|b_n| \leq a_n \leq |b_n|$.
8. $\frac{1}{n} \sum_{k \leq n} f(k/n)$: that is a Riemann sum, see §9.6.
9. **Nothing fits:** treat n as a real x and use Taylor / l’Hospital (§5.3, §8.5).

FACT: tools with names — cite them, it earns marks

Limit laws: sums, products, quotients (denominator $\neq 0$) — only if *both* limits exist finitely.
Squeeze (Sandwich): $b_n \leq a_n \leq c_n$ and $b_n, c_n \rightarrow L \Rightarrow a_n \rightarrow L$.
Monotone convergence (Monotoniekriterium): monotone + bounded $\Rightarrow$ convergent, limit = sup resp. inf of the term set.
Continuity transfer: $a_n \rightarrow a$ and f continuous at $a \Rightarrow f(a_n) \rightarrow f(a)$.
Ratio criterion for sequences: $\left| \frac{a_{n+1}}{a_n} \right| \rightarrow q < 1 \Rightarrow a_n \rightarrow 0$ geometrically.
! Do not look for Leibniz / Quotienten- / Wurzel- / Majorantenkriterium here: those decide whether a *series* $\sum a_n$ converges, and they all live in §4.2–§4.3.

EXAMPLE: the limits that keep coming back

- $\frac{3n^2-n}{2n^2+5} = \frac{3-\frac{1}{n}}{2+\frac{5}{n^2}} \rightarrow \frac{3}{2}$ (divide by n^2).
- **Conjugate, in full:** $\sqrt{n^2+n} - n = \frac{(\sqrt{n^2+n}-n)(\sqrt{n^2+n}+n)}{\sqrt{n^2+n}+n} = \frac{(n^2+n)-n^2}{\sqrt{n^2+n}+n} = \frac{n}{\sqrt{n^2+n}+n} = \frac{1}{\sqrt{1+\frac{1}{n}}+1} \rightarrow \frac{1}{2}$.
(last step: divide by n .)
- $\sqrt[n]{n} = e^{\frac{\ln n}{n}} \rightarrow 1$; $\sqrt[n]{a} \rightarrow 1$ ($a > 0$); $\sqrt[n]{2n+3} \rightarrow 3$ (pull out the biggest).
- $\left(1 + \frac{3}{n}\right)^{2n} = \left[\left(1 + \frac{3}{n}\right)^n\right]^2 \rightarrow e^6$.
- $\frac{n!}{n^n} \leq \frac{1}{n} \rightarrow 0$; $\frac{2^n}{n!} \rightarrow 0$ (ratio $\frac{2^{n+1}}{n+1} \leq \frac{1}{2}$ for $n \geq 3$).
- $\frac{(-1)^n}{n}$: $|a_n| \leq \frac{1}{n} \rightarrow 0 \Rightarrow 0$. Alternating by itself means nothing.

EXAMPLE: two exam-style limits (D-INFK FS25 2.A2)

Bounded inside a continuous function: $a_n = \tan\left(\frac{\sin a_n}{4\sqrt{n}}\right)$. $|\sin n| \leq 1 \Rightarrow \left|\frac{\sin n}{4\sqrt{n}}\right| \leq \frac{1}{4\sqrt{n}} \rightarrow 0$ (squeeze); $\tan$ is continuous at $0 \Rightarrow a_n \rightarrow \tan 0 = 0$.
log of a sum: $b_n = \frac{\log(n+e^{n^2})}{2n^2}$. Factor out the dominant term: $\log(e^{n^2}(1 + ne^{-n^2})) = n^2 + \underbrace{\log(1 + ne^{-n^2})}_{\rightarrow 0} \Rightarrow b_n \rightarrow \frac{n^2}{2n^2} = \frac{1}{2}$.

► RECIPE: prove $a_n \rightarrow a$ by ϵ - N

1. Compute $|a_n - a|$, simplify to one fraction.
2. Over-estimate to a clean $\frac{C}{n^k}$ (numerator bigger, denominator smaller).
3. Solve $\frac{C}{n^k} < \epsilon \Rightarrow$ take $N := \lceil (C/\epsilon)^{1/k} \rceil + 1$.
4. Write it out: “Let $\epsilon > 0$, choose N as above; then $\forall n \geq N: |a_n - a| \leq \frac{C}{n^k} < \epsilon$.” $\square$

FACT: estimates worth remembering

$n! \geq 2^{n-1}$; $(\frac{n}{3})^n \leq n! \leq n^n$; **Stirling** $n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$, so $\sqrt[n]{n!} \sim \frac{n}{e} \rightarrow \infty$.
 $\sum_{k=1}^n \frac{1}{k} \approx \ln n + \gamma$ (grows, unbounded); $\left(1 + \frac{1}{n}\right)^n \uparrow e$, $\left(1 + \frac{1}{n}\right)^{n+1} \downarrow e$.

3.3 Recursive sequences $x_{n+1} = f(x_n)$

► RECIPE: “konvergiert (x_n) , und wogegen?” — the full treatment (4_9 -P)

The task: you are given x_0 and a rule $x_{n+1} = f(x_n)$, never a formula for x_n . ! You may **not** just solve $L = f(L)$: that equation only says *if* the limit exists, it is one of these roots. Steps 2–3 are what earn you the right to take the limit at all.

1. **Candidate limits:** solve $L = f(L)$. Keep *every* root — step 4 decides which one.
2. **Bounded:** find an interval I with $x_0 \in I$ and $f(I) \subseteq I$ (“ I is invariant”); induction then gives $x_n \in I$ for all n .
3. **Monotone:** look at the sign of $f(x) - x$ on I — positive means $x_{n+1} > x_n$, so increasing; negative means decreasing. If f is increasing, that sign can never flip, so the direction holds for *all* n .
4. **Conclusion:** monotone + bounded $\Rightarrow$ converges (§3.2). Only now write $L = \lim x_{n+1} = \lim f(x_n) = f(L)$, using continuity of f ; pick the root of $L = f(L)$ that lies in I .

FACT: contraction shortcut + DEF: what q is (often much faster)

q is a **contraction factor**: any constant with $|f'(x)| \leq q < 1$ for **all** $x \in I$ — so f shrinks every distance by at least the factor q . Get it by maximising $|f'|$ on I . ! $|f'| < 1$ pointwise is not enough; you need one q bounded away from 1.
Then by the MVT $|x_{n+1} - L| = |f(x_n) - f(L)| \leq q|x_n - L|$, so

$$|x_n - L| \leq q^n |x_0 - L| \rightarrow 0 \quad \text{geometrically.}$$

$\Rightarrow$ exactly **one** fixed point in I , and x_n converges to it from *every* start $x_0 \in I$. **No monotonicity and no boundedness argument needed** — this replaces steps 2–4 above.

EXAMPLE: $x_{n+1} = \frac{\sin x_n}{1+x_n^2} + x_0 \in (0, 1)$

$f(x) = \frac{\sin x}{1+x^2}$. 1) $f(0) = 0$; for $x \neq 0$, $|\sin x| < |x|$ and $1+x^2 \geq 1$ give $|f(x)| < |x| \Rightarrow$ no other fixed point. 2) $x_0 \in (0, 1) \Rightarrow \sin x_0 > 0 \Rightarrow$ all $x_n > 0$, and $x_{n+1} < x_n \Rightarrow$ strictly decreasing, bounded below by 0. 3) Monotone + bounded $\Rightarrow$ converges to $L \geq 0$ with $L = f(L) \Rightarrow L = 0$.

3.4 Accumulation points, limsup, liminf

DEF: accumulation point (Häufungspunkt)

h is an **accumulation point** of $(a_n) \iff$ every ϵ -ball around h contains a_n for **infinitely many** $n \iff$ **some subsequence converges** to h . The second form is the one you use.
! A *limit* is where the whole sequence goes; an *accumulation point* is where some part of it keeps returning. A sequence has at most **one** limit but may have many accumulation points.

► RECIPE: find all accumulation points

Idea: the sequence has one ingredient that **repeats**. Split n along its period; each residue class is a subsequence with an honest limit, and those limits are the answer.

1. Find the repeating ingredient and split n accordingly:
 - $(-1)^n$ has **period 2** $\Rightarrow$ two cases $n = 2k$ (gives +1) and $n = 2k + 1$ (gives -1).
 - $\cos \frac{n\pi}{3}$ has **period 6** $\Rightarrow$ six cases $n \equiv 0, \dots, 5 \pmod{6}$.
 - $\sin n$ **never repeats** (as π is irrational) $\Rightarrow$ the values are *dense*, so **every** point of $[-1, 1]$ is an accumulation point.
2. Take the limit of each subsequence separately. That set of limits is **exactly** the set of accumulation points.
3. $\limsup = \max$ of that set, $\liminf = \min$ of that set.

Ex. $a_n = (-1)^n \frac{n}{n+1}$: $n = 2k \Rightarrow \frac{2k}{2k+1} \rightarrow +1$; $n = 2k + 1 \Rightarrow -\frac{2k+1}{2k+2} \rightarrow -1$. Set $\{-1, +1\}$, so $\limsup = 1$, $\liminf = -1$, and since they differ the sequence **diverges**.

FACT: limsup / liminf algebra + DEF and intuition

Intuition: $\limsup a_n$ is the **largest accumulation point** — the highest level the sequence keeps climbing back to; $\liminf a_n$ is the **smallest**. Formally $\limsup a_n = \lim_{n \rightarrow \infty} \sup_{k \geq n} a_k$ (the running sup can only fall, so the limit exists), $\liminf = \lim_n \inf_{k \geq n} a_k$.
Always defined in $[-\infty, \infty]$, no hypotheses needed. $\liminf a_n \leq \limsup a_n$.

- a_n converges $\iff$ bounded and $\liminf = \limsup$ (then all three coincide).
- $\limsup(a_n + b_n) \leq \limsup a_n + \limsup b_n$ — ! only $\leq$, equality can fail ($a_n = (-1)^n$, $b_n = -(-1)^n$).
- $\liminf(a_n + b_n) \geq \liminf a_n + \liminf b_n$; $\liminf(-a_n) = -\limsup a_n$.
- If $b_n \rightarrow b$ exists then $\limsup(a_n + b_n) = \limsup a_n + b$ (a convergent summand splits off exactly).

TRAP: monotone sequences and accumulation points

A monotone sequence is either bounded ($\Rightarrow$ converges, **exactly one** accumulation point) or unbounded ($\Rightarrow \rightarrow \pm\infty$, **no** accumulation point).
So “a monotone sequence may have no accumulation point” T, while “at least one” F and “at least two” F. Bolzano–Weierstrass only promises one for *bounded* sequences.

FACT: nested intervals (Intervallschachtelung) + DEF

A nested interval sequence is a chain $I_1 \supseteq I_2 \supseteq \dots$ of **closed, bounded** intervals whose lengths satisfy $|I_n| \rightarrow 0$.
 $\Rightarrow \bigcap_n I_n$ is **exactly one point**. ! All three hypotheses are needed: $(0, \frac{1}{n})$ (not closed) and $[n, \infty)$ (not bounded) both have empty intersection.
Equivalent to completeness; it is the engine of the bisection proof of Bolzano–Weierstrass (§11).

3.5 Cauchy sequences

DEF: Cauchy

$\forall \epsilon > 0 \exists N \forall m, n \geq N: |a_n - a_m| < \epsilon$. In $\mathbb{R}$: **Cauchy $\iff$ convergent**, so this proves convergence *without knowing the limit*.

TRAP: the near-miss

“ $|a_{n+1} - a_n| \rightarrow 0$ ” is **not** enough: $a_n = \sqrt{n}$ and $a_n = \sum_{k \leq n} \frac{1}{k}$ both satisfy it and diverge. You need control of $|a_n - a_m|$ for *all* $m, n \geq N$, not just neighbours.

► RECIPE: show (a_n) is Cauchy + DEF of the two constants

Goal: bound the gap between *consecutive* terms by a geometric sequence, then sum it.

1. Find constants $C > 0$ (any prefactor — whatever the first estimate throws up) and $0 < q < 1$ (the geometric rate) with $|a_{n+1} - a_n| \leq C q^n$ for all n . A recursion with a contraction factor q (§3.3) hands you exactly this.
2. Telescope for $m > n$: $|a_m - a_n| \leq \sum_{k=n}^{m-1} |a_{k+1} - a_k| \leq \sum_{k=n}^{m-1} C q^k \leq \sum_{k=n}^{\infty} C q^k \leq \frac{C q^n}{1-q}$ (geometric series, §4.2).
3. That bound depends on n alone and $\rightarrow 0$, so given ϵ choose N with $\frac{C q^N}{1-q} < \epsilon \Rightarrow$ **Cauchy** $\Rightarrow$ convergent, without ever naming the limit.

4 Series $\sum a_n$
4.1 Setup
DEF: series, absolute vs conditional $\sum_{n=1}^{\infty} a_n := \lim_{N \rightarrow \infty} s_N$ with $s_N = \sum_{n=1}^N a_n$ — a series is the sequence of partial sums. absolutely convergent: $\sum a_n < \infty$. abs. conv. $\Rightarrow$ conv., not conversely ($\sum \frac{(-1)^n}{n}$). conditionally convergent: converges but not absolutely. Convergence is unaffected by finitely many terms or by the starting index.
4.2 Decision tree: does $\sum a_n$ converge?
► RECIPE: run these in order, stop at the first hit 1. $a_n \not\rightarrow 0$? $\Rightarrow$ diverges, done (<i>Trivialekriterium</i>). Free, always check first. 2. Known shape? geometric, p-series, log-p — read the answer off the box below. 3. Factorials or c^n? $\Rightarrow$ ratio test. 4. n-th powers, n in the exponent? $\Rightarrow$ root test (stronger than ratio). 5. Resembles a known series? $\Rightarrow$ comparison on leading powers. 6. Alternating? $\Rightarrow$ Leibniz. 7. $a_n = f(n)$ with $f \geq 0$ decreasing? $\Rightarrow$ integral test. 8. Telescoping? $a_n = b_n - b_{n+1} \Rightarrow s_N = b_1 - b_{N+1}$. ! With a <i>shift</i> of k ($a_n = b_n - b_{n+k}$) exactly k terms survive at each end: $\sum_{n \geq 2} (\frac{1}{n+1} - \frac{1}{n-1})$ leaves $-\frac{1}{1} - \frac{1}{2}$.
FACT: the benchmark series (memorise these four lines) $\sum_{n \geq 0} q^n$ conv. $\iff q < 1$, value $\frac{1}{1-q}$; $\sum_{n \geq 1} nq^{n-1} = \frac{1}{(1-q)^2}$. $\sum \frac{1}{n^p}$ conv. $\iff p > 1$. ($p = 1$ harmonic, diverges.) $\sum_{n \geq 2} \frac{1}{n \ln^n n}$ conv. $\iff p > 1$. $\sum \frac{1}{n!} = e$; $\sum_{n \geq 1} \frac{(-1)^{n+1}}{n} = \ln 2$; $\sum \frac{1}{n^2} = \frac{\pi^2}{6}$; $\sum \frac{1}{n(n+1)} = 1$.
FACT: ratio and root test, stated exactly Ratio (<i>Quotientenkriterium</i>): $q = \lim \left \frac{a_{n+1}}{a_n} \right$. $q < 1$ abs. conv.; $q > 1$ div.; $q = 1$ no information. Root (<i>Wurzelkriterium</i>): $q = \limsup \sqrt[n]{ a_n }$, same verdicts. ! Both give $q = 1$ for $\sum \frac{1}{n}$ and $\sum \frac{1}{n^p}$, which behave differently — at $q = 1$ you must switch method.
FACT: comparison, Leibniz, integral test Comparison (<i>Majorante/Minorante</i>): $0 \leq a_n \leq b_n$, $\sum b_n < \infty \Rightarrow \sum a_n < \infty$; $a_n \geq b_n \geq 0$, $\sum b_n = \infty \Rightarrow \sum a_n = \infty$. Compare leading powers only: $\frac{3n^2+n}{n^3-1} \sim \frac{3}{n^2} \Rightarrow$ conv. Limit comparison: $\frac{a_n}{b_n} \rightarrow c \in (0, \infty) \Rightarrow$ same behaviour. Leibniz: $a_n = (-1)^n b_n$ with $b_n \geq 0$ monotonically decreasing to 0 $\Rightarrow$ converges, and $s - s_N \leq b_{N+1}$. Integral test: $f \geq 0$ decreasing $\Rightarrow \int_1^\infty f dx \leq \sum_{n \geq 1} f(n) \leq f(1) + \int_1^\infty f dx$; sum and integral converge together.
TRAP: test traps • Leibniz needs <i>monotone</i> b_n: $b_n = \frac{2+(-1)^n}{n} \rightarrow 0$ but is not monotone. • Comparison needs $a_n \geq 0$; with signs, apply it to a_n (that proves <i>absolute</i> convergence only). • $\sum a_n$ conv. and $\sum b_n$ conv. $\Rightarrow \sum (a_n + b_n)$ conv. T, but $\sum a_n b_n$ conv. F ($a_n = b_n = \frac{(-1)^n}{\sqrt{n}}$). • “$s_N$ bounded $\Rightarrow$ converges” F unless $a_n \geq 0$ (then s_N is monotone).
EXAMPLE: run the tree on five series $\sum \frac{n^2}{2^n}$: $c^n \Rightarrow$ ratio $\frac{(n+1)^2}{2n^2} \rightarrow \frac{1}{2} < 1 \Rightarrow$ conv. $\sum \frac{n!}{n^n}$: ratio $= (\frac{n}{n+1})^n \rightarrow \frac{1}{e} < 1 \Rightarrow$ conv. $\sum (\frac{n}{2n+1})^n$: root $= \frac{n}{2n+1} \rightarrow \frac{1}{2} < 1 \Rightarrow$ conv. $\sum \frac{2n+3}{n^2+7}$: behaves like $\frac{2}{n} \Rightarrow$ div. $\sum \frac{(-1)^n}{\sqrt{n}}$: Leibniz $\Rightarrow$ conv.; $\sum \frac{1}{\sqrt{n}}$ div. $\Rightarrow$ conditionally conv.
EXAMPLE: telescoping via partial fractions $\sum_{n \geq 1} \frac{1}{n(n+2)}$: split $\frac{1}{n(n+2)} = \frac{1}{2} \left(\frac{1}{n} - \frac{1}{n+2} \right)$. $s_N = \frac{1}{2} \left(1 + \frac{1}{2} - \frac{1}{N+1} - \frac{1}{N+2} \right) \rightarrow \frac{3}{4}$. Recipe: partial-fraction it (how: §9.4), write out the first three and last two terms, see what survives.

4.3 Rearrangement, products, mixed-sign tests
FACT: rearrangement and Cauchy product • Riemann rearrangement: conditionally convergent $\Rightarrow$ a reordering can be made to converge to <i>any</i> prescribed value, or to $\pm\infty$. Order matters. • Dirichlet: absolutely convergent $\Rightarrow$ every reordering gives the same sum. • Cauchy product $c_n = \sum_{k=0}^n a_k b_{n-k}$: if at least one factor converges absolutely then $\sum c_n = (\sum a_n)(\sum b_n)$ (Mertens).
FACT: Abel / Dirichlet test (for $\sum a_n b_n$ with signs) If $b_n \downarrow 0$ monotonically and the partial sums of $\sum a_n$ are <i>bounded</i>, then $\sum a_n b_n$ converges. This is what makes $\sum \frac{\sin n}{n}$ and $\sum \frac{\cos n}{n}$ converge although neither is absolutely convergent, and it is the discrete twin of the integration-by-parts argument for $\int_1^\infty \frac{\cos t}{t} dt$.

5

Power series and Taylor series

5.1 Radius of convergence
► RECIPE: “for which x does $\sum a_n (x - x_0)^n$ converge?” 1. Read off the centre x_0 (in $(x - 2)^n$: $x_0 = 2$) and the coefficient $a_n =$ everything without $(x - x_0)^n$. 2. Radius R — both formulas always give the same R, so pick whichever is easier to evaluate: ratio $\frac{1}{R} = \lim \left \frac{a_{n+1}}{a_n} \right$ when a_n is built from factorials or products (they cancel); root (<i>Cauchy–Hadamard</i>) $\frac{1}{R} = \limsup \sqrt[n]{ a_n }$ when n sits in an exponent, and the only one that is always valid. Limit $0 \Rightarrow R = \infty$ (converges everywhere); limit $\infty \Rightarrow R = 0$ (only at x_0). 3. ! Only even/odd powers ($\sum a_n x^{2n}$)? Substitute $u = x^2$, find the radius R_u in u, then $x < \sqrt{R_u}$. Reading R straight off a_n gives the wrong interval. 4. Absolute convergence on $x - x_0 < R$, divergence on $x - x_0 > R$. 5. Endpoints $x = x_0 \pm R$ one at a time — plug in and use §4.2 (usually Leibniz or p-series). The two ends can behave differently. 6. Answer as an interval, e.g. $[x_0 - R, x_0 + R)$.
EXAMPLE: $\sum_{n \geq 1} \frac{n^n (x-2)^n}{(2n)!}$ $a_n = \frac{n^n}{(2n)!}$; factorial $\Rightarrow$ ratio test. $\left \frac{a_{n+1}}{a_n} \right = \frac{(n+1)^{n+1}}{(2n+2)!} \cdot \frac{(2n)!}{n^n} = \frac{(n+1)}{(2n+1)(2n+2)} \left(1 + \frac{1}{n} \right)^n \longrightarrow 0,$ since the numerator grows like $e n$ and the denominator like $4n^2$. So $\frac{1}{R} = 0$, $R = \infty$: converges for every $x \in \mathbb{R}$.
FACT: what a power series gives you for free On $x - x_0 < R$ the sum is C^∞ and you may differentiate and integrate termwise; R is unchanged. Coefficients are unique: $a_n = \frac{f^{(n)}(x_0)}{n!}$. Convergence is <i>uniform</i> on every closed $[x_0 - r, x_0 + r]$ with $r < R$ (Weierstrass M-test), ! not necessarily on the whole open disc.

5.2 Taylor formula and error bounds
DEF: linear / tangent approximation $f(x) \approx f(x_0) + f'(x_0)(x - x_0)$ — the $n = 1$ Taylor polynomial, i.e. the tangent line. Error $\leq \frac{\max f'' }{2} (x - x_0)^2$. This is also the <i>definition</i> of differentiability: the tangent is the unique linear function whose error is $o(x - x_0)$.
FACT: Taylor with Lagrange remainder $f(x) = \underbrace{\sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k}_{T_n(x)} + R_n(x), \quad R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)^{n+1} \text{ for some } \xi \text{ between } x_0 \text{ and } x;$ also $R_n = o(x - x_0 ^n)$. Error bound: $R_n(x) \leq \frac{M}{(n+1)!} x - x_0 ^{n+1}$ with $M = \max f^{(n+1)}$ on the interval between x_0 and x.
► RECIPE: approximate a value and bound the error 1. Pick x_0 near x where f and its derivatives are known exactly $(0, 1, \frac{\pi}{6}, \dots)$. 2. Write T_n and evaluate. 3. Bound $M = \max f^{(n+1)}$ on the interval (for sin, cos: $M = 1$ always; for e^x: the right endpoint). 4. If an accuracy ϵ is prescribed, solve $\frac{M}{(n+1)!} x - x_0 ^{n+1} < \epsilon$ for n. Ex. $e^{0.1}$, $n = 3$: $T_3 = 1 + 0.1 + 0.005 + \frac{0.001}{6} \approx 1.10517$; $M < 1.2 \Rightarrow R_3 \leq \frac{1.2}{24} 10^{-4} = 5 \cdot 10^{-6}$.

5.3 Limits by Taylor: the fastest tool you have
► RECIPE: limits $x \rightarrow 0$ by plugging in series 1. Look at the denominator's power x^m. You need every series to order m — one order further if the leading terms cancel. 2. Insert the standard series (§13.3), expand, cancel. 3. The surviving lowest-order term is the answer; everything else is $o(x^m)$. $O(x^k)$ means “a rest that is at most a constant times x^k near 0” — a bin for everything too small to matter (full rules §8.1). Bookkeeping: $\sin x = x + O(x^3)$, so $\frac{\sin x}{x} = 1 + O(x^2)$. Carry the $O(x^k)$ through every step and you cannot lose a term.
EXAMPLE: $\lim_{x \rightarrow 0} \frac{\sin x - x + \frac{x^3}{6}}{x^5}$ $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040} + \dots$ Numerator $= \frac{x^5}{120} + O(x^7) \Rightarrow$ limit $= \frac{1}{120}$. l'Hospital would need five differentiations. Don't.
EXAMPLE: more Taylor limits, one line each $\frac{1 - \cos x}{x^2} = \frac{x^2/2 + O(x^4)}{x^2} \rightarrow \frac{1}{2}$. $\frac{e^x - 1 - x}{x^2} = \frac{x^2/2 + O(x^3)}{x^2} \rightarrow \frac{1}{2}$. $\frac{\tan x - x}{x^3} \rightarrow \frac{1}{3}$ (since $\tan x = x + \frac{x^3}{3} + O(x^5)$). $\frac{1}{\sin x} - \frac{1}{x} = \frac{x - \sin x}{x^2 \sin x} = \frac{x^3/6}{x^2} \rightarrow 0$. $\frac{\ln(1+x) - x}{x^2} \rightarrow -\frac{1}{2}$.
5.4 Building new series from old ones
► RECIPE: never recompute derivatives — transform instead 1. Substitute: $e^{-x^2} = \sum \frac{(-1)^n x^{2n}}{n!}$; $\frac{1}{1+x^2} = \sum (-1)^n x^{2n}$. 2. Differentiate termwise: $\frac{1}{(1-x)^2} = \sum_{n \geq 1} n x^{n-1}$. 3. Integrate termwise: $\arctan x = \int \frac{-dx}{1+x^2} = \sum \frac{(-1)^n x^{2n+1}}{2n+1}$; $\ln(1+x) = \int \frac{dx}{1+x}$. 4. Multiply: collect by total degree (Cauchy product). $e^x \sin x = x + x^2 + \frac{x^3}{3} + O(x^4)$. 5. Divide: never do long division. Write the denominator as $1 - u$ (pull out its constant term first) and use $\frac{1}{1-u} = 1 + u + u^2 + \dots$, then multiply. Ex. $\tan x = \frac{\sin x}{\cos x} = \cos x - \left(\frac{x^2}{2} + O(x^4) \right)$, so $\frac{1}{\cos x} = 1 + \frac{x^2}{2} + O(x^4)$, and $\tan x = \left(x - \frac{x^3}{6} \right) \left(1 + \frac{x^2}{2} \right) + O(x^5) = x + \frac{x^3}{3} + O(x^5)$. 6. Shift centre: to expand at $x_0 \neq 0$, substitute $t = x - x_0$ and expand in t.
FACT: ready-made compositions $\ln(1+x^2) = x^2 - \frac{x^4}{2} + \frac{x^6}{3} - \dots$; $\sin(x^2) = x^2 - \frac{x^6}{6} + \dots$; $e^{-x^2} = 1 - x^2 + \frac{x^4}{2} - \dots$; $\frac{1}{1+x^2} = 1 - x^2 + x^4 - \dots$; $\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} - \dots$; $\frac{1}{\sqrt{1-x}} = 1 + \frac{x}{2} + \frac{3x^2}{8} + \dots$
6 Functions: limits, continuity, IVT/EVT
6.1 Basic properties, inverse
► RECIPE: injective / surjective / bijective / inverse 1. Injective: $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$. Shortcut: <i>strictly</i> monotone $\Rightarrow$ injective (show $f' > 0$ or $f' < 0$ throughout). 2. Surjective onto Y: solve $f(x) = y$ for arbitrary $y \in Y$, or get $\text{im } f = Y$ from IVT plus the limits at the ends of the domain. 3. Bijective = both $\Rightarrow f^{-1}$ exists; compute it by solving $y = f(x)$ for x, then swap names. 4. Regularity of f^{-1}: f continuous and strictly monotone on an interval $\Rightarrow f^{-1}$ continuous and strictly monotone; and differentiable with $(f^{-1})'(y_0) = \frac{1}{f'(x_0)}$ whenever $f'(x_0) \neq 0$. Also: even $f(-x) = f(x)$, odd $f(-x) = -f(x)$, periodic $f(x+T) = f(x)$, bounded $f \leq C$, restriction $f _A$.
FACT: monotone functions are well behaved Monotone on an interval $\Rightarrow$ at most <i>countably many</i> discontinuities, all of them jumps; both one-sided limits exist everywhere; Riemann integrable on every $[a, b]$. It can still fail to be continuous, and it is injective only if <i>strictly</i> monotone.

6.2 Limits of functions

DEF: ε - δ , one-sided, at infinity

$$\lim_{x \rightarrow x_0} f(x) = L : \iff \forall \varepsilon > 0 \exists \delta > 0 : 0 < |x - x_0| < \delta \implies |f(x) - L| < \varepsilon.$$

Limit exists $\iff \lim_{x \rightarrow x_0^-} f = \lim_{x \rightarrow x_0^+} f$, both finite.
 $\lim_{x \rightarrow \infty} f = L : \iff \forall \varepsilon \exists K : x > K \implies |f(x) - L| < \varepsilon.$

► RECIPE: compute $\lim_{x \rightarrow x_0} f(x)$

1. Plug in x_0 . Nothing breaks $\implies$ done, by continuity.
2. $\frac{0}{0} \implies$ factor and cancel, expand roots by the conjugate, else Taylor (§5.3) or l'Hospital (§8.5).
3. $\frac{\pm\infty}{0} \implies \pm\infty$; get the sign from each side separately, so the answer may exist only one-sided.
4. $x \rightarrow \pm\infty$ rational $\implies$ divide by the highest power in the denominator. ! for $x \rightarrow -\infty$, $\sqrt{x^2} = |x| = -x$.
5. $|x|$, sgn, floor, piecewise $\implies$ **always** both one-sided limits.
6. Bounded $\times$ null $\implies$ squeeze: $x \sin \frac{1}{x} \rightarrow 0$, while $\sin \frac{1}{x}$ has no limit at all.

6.3 Continuity

DEF: continuity and the sequential criterion

$$f \text{ continuous at } x_0 : \iff \forall \varepsilon > 0 \exists \delta > 0 : |x - x_0| < \delta \implies |f(x) - f(x_0)| < \varepsilon$$
$$\iff \lim_{x \rightarrow x_0} f(x) = f(x_0) \iff (\text{Heine}) x_n \rightarrow x_0 \text{ implies } f(x_n) \rightarrow f(x_0) \text{ for every such sequence.}$$

► RECIPE: prove or disprove continuity at x_0

Prove: if f is built from $+$, $-$, $\times$, $\div$ (denominator $\neq 0$) and $\circ$ of continuous standard functions, say exactly that — one line. Otherwise ε - δ : bound $|f(x) - f(x_0)| \leq C|x - x_0|$ and take $\delta = \varepsilon/C$.
Disprove: give two sequences $x_n \rightarrow x_0$, $y_n \rightarrow x_0$ with $\lim f(x_n) \neq \lim f(y_n)$ (Heine). Classics: $x_n = \frac{1}{2\pi n}$ vs $y_n = \frac{1}{\frac{\pi}{2} + 2\pi n}$ for $\sin \frac{1}{x}$; rationals vs irrationals for Dirichlet.

FACT: types of discontinuity (Unstetigkeitsstellen)

removable (hebbar): both one-sided limits exist and agree, but $\neq f(x_0)$ — redefine and it is continuous.
jump (Sprung): both exist and differ. This is the monotone-function case.
essential / oscillating: at least one one-sided limit fails to exist ($\sin \frac{1}{x}$, $\frac{1}{x}$).

► RECIPE: fit a parameter: “determine a, b so that f is continuous / differentiable”

1. Continuity at the seam x_0 : $\lim_{x \rightarrow x_0^-} f = \lim_{x \rightarrow x_0^+} f = f(x_0) \implies$ one equation.
2. Differentiability: additionally $\lim_{x \rightarrow x_0^-} f' = \lim_{x \rightarrow x_0^+} f' \implies$ second equation.
3. Solve the (usually linear) system. Two unknowns need exactly these two conditions.

6.4 The three theorems on $[a, b]$ — and when to reach for which

FACT: cite these verbatim, hypotheses included

$f : [a, b] \rightarrow \mathbb{R}$ **continuous**, $[a, b]$ **compact** (closed and bounded, §1.1):

- IVT (Zwischenwertsatz): c between $f(a)$ and $f(b) \implies \exists x \in [a, b] : f(x) = c$. For: *a zero, a solution, a fixed point exists.*
- Extreme value theorem (Satz vom Max/Min, Weierstrass): f is bounded and *attains* max and min. For: *“the maximum is attained”, bounding.*
- Heine: continuous on a compact interval $\implies$ *uniformly* continuous. For: *uniform continuity for free.*

! All three fail on open or unbounded intervals: $\frac{1}{x}$ on $(0, 1]$ is continuous, unbounded, attains no max, is not uniformly continuous.

► RECIPE: show “ $f(x) = g(x)$ has a solution” / “ f has a zero”

1. Set $h := f - g$ (fixed point $f(x) = x$: take $h(x) := f(x) - x$). State: h continuous as a difference of continuous functions.
2. Find two points of opposite sign — evaluate at convenient points, or use the signs of $\lim_{x \rightarrow \pm\infty} h$ and pick large $|x|$.
3. IVT $\implies \exists x^*$ with $h(x^*) = 0$. Existence done.
4. Uniqueness: h strictly monotone ($h' > 0$ or $h' < 0$ everywhere) $\implies$ at most one zero. Or Rolle: two zeros would force $h'(\xi) = 0$, contradiction.

EXAMPLE: $e^x = p(x)$ — the standard multiple choice

- “IVT always gives a solution” **F** — $p \equiv -1$ has none, since $e^x > 0$.
- “ p of odd degree $\implies$ always at least one solution” **T** — $e^x - p(x)$ changes sign at $\pm\infty$.
- “at most finitely many solutions for every p ” **T** — $e^x - p(x)$ is analytic and not identically 0, so its zeros are isolated and finite in number on $\mathbb{R}$ here.
- “for every N there is a p_N with $\geq N$ solutions” **T** — take p_N close to e^x on a long interval and let it wobble across.

6.5 Uniform continuity (gleichmässige Stetigkeit)

DEF: the difference that matters

continuous: $\forall x_0 \forall \varepsilon \exists \delta(\varepsilon, x_0)$. **uniformly continuous:** $\forall \varepsilon \exists \delta(\varepsilon) \forall x, y : |x - y| < \delta \implies |f(x) - f(y)| < \varepsilon$.

One δ for the whole domain. unif. cont. $\implies$ cont., never the converse.
Chain: Lipschitz $\implies$ uniformly continuous $\implies$ continuous.

► RECIPE: “is f uniformly continuous?”

1. Compact domain $[a, b]$, f continuous? $\implies$ YES by Heine. One line, done.
2. f' bounded, $|f'| \leq L$? $\implies$ MVT gives $|f(x) - f(y)| \leq L|x - y|$, Lipschitz $\implies$ YES with $\delta = \varepsilon/L$. The workhorse on unbounded domains.
3. Continuous on $\mathbb{R}$ with finite limits at $\pm\infty$? $\implies$ YES (compact in the middle, nearly constant outside).
4. Suspect NO? Produce x_n, y_n with $|x_n - y_n| \rightarrow 0$ but $|f(x_n) - f(y_n)| \geq \varepsilon_0 > 0$.
5. Parameter that can degenerate (e.g. $n = 0$ vs $n \geq 1$) $\implies$ case distinction, treat the degenerate case on its own.

! Unbounded f' does **not** prove failure: $\sqrt{x}$ on $[0, 1]$ has $f' \rightarrow \infty$ and is uniformly continuous (compact).

EXAMPLE: uniformly continuous: $f_n(x) = \frac{x}{1 + nx^2}$ on $\mathbb{R}$

$n = 0$: $f_0(x) = x$ is Lipschitz with $L = 1$.
 $n \geq 1$: $f'_n(x) = \frac{1 - nx^2}{(1 + nx^2)^2}$, so $|f'_n(x)| \leq \frac{1 + nx^2}{(1 + nx^2)^2} = \frac{1}{1 + nx^2} \leq 1$.
Bounded derivative $\implies$ Lipschitz with $L = 1 \implies$ **uniformly continuous for every n , with $\delta = \varepsilon$.**

EXAMPLE: not uniformly continuous: the two patterns

$f(x) = x^2$ on $\mathbb{R}$: $x_n = n$, $y_n = n + \frac{1}{n} \implies |x_n - y_n| = \frac{1}{n} \rightarrow 0$ but $|f(y_n) - f(x_n)| = 2 + \frac{1}{n^2} \geq 2 \implies$ fails for $\varepsilon = 2$.
 $f(x) = \frac{1}{x}$ on $(0, 1)$: $x_n = \frac{1}{n}$, $y_n = \frac{1}{2n} \implies$ difference $\rightarrow 0$ but $|f(y_n) - f(x_n)| = n \rightarrow \infty$.
 $\sin \frac{1}{x}$ on $(0, 1)$: pick x_n, y_n at consecutive max/min of the oscillation $\implies$ gap 2.

7 Sequences of functions (f_n)

7.1 Pointwise limit: always compute this first

DEF: pointwise vs uniform, side by side

pointwise: $\forall x \forall \varepsilon \exists N(\varepsilon, x) \forall n \geq N : |f_n(x) - f(x)| < \varepsilon$ — N may depend on x .
uniform: $\forall \varepsilon \exists N(\varepsilon) \forall n \geq N \forall x : |f_n(x) - f(x)| < \varepsilon$, i.e.

$$\|f_n - f\|_\infty := \sup_{x \in D} |f_n(x) - f(x)| \xrightarrow{n \rightarrow \infty} 0.$$

One N for all x at once. Uniform $\implies$ pointwise, never the converse.

► RECIPE: find the pointwise limit $f(x) = \lim_n f_n(x)$

1. Freeze x , treat it as a constant, let $n \rightarrow \infty$ with §3.2.
2. Case distinction in x wherever the answer changes: usually $x = 0$ vs $x \neq 0$, or $|x| < 1$ vs $|x| = 1$ vs $|x| > 1$.
3. Write f piecewise. If f is discontinuous while all f_n are continuous, convergence cannot be uniform — that alone answers part (c).

7.2 Uniform convergence $f_n \rightrightarrows f$

► RECIPE: decide uniform convergence (the standard 3-point task)

1. Get the pointwise limit f first — uniform convergence can only be towards it.
2. Set $g_n(x) := |f_n(x) - f(x)|$ and compute $M_n := \sup_x g_n(x)$: differentiate in x with n fixed, solve $g'_n(x) = 0$, substitute the critical point back. Check the boundary and $|x| \rightarrow \infty$ too.
3. $M_n \rightarrow 0 \implies$ **uniform**. $M_n \not\rightarrow 0 \implies$ **pointwise only**.

► RECIPE: two shortcuts that give a complete disproof

- Exhibit one sequence x_n with $|f_n(x_n) - f(x_n)| \geq \varepsilon_0 > 0$ (take x_n = the maximiser). One such sequence is a full proof of failure — you do not need the exact sup.
- All f_n continuous but f is not $\implies$ not uniform, by the contrapositive of the uniform limit theorem.

EXAMPLE: $f_n(x) = \frac{x}{1 + nx^2}$ on $\mathbb{R}$: uniform

Pointwise $f \equiv 0$, so $M_n = \sup_x \left| \frac{x}{1 + nx^2} \right|$.
 $f'_n(x) = \frac{1 - nx^2}{(1 + nx^2)^2} = 0 \implies x = \pm \frac{1}{\sqrt{n}}$, and $f_n\left(\frac{1}{\sqrt{n}}\right) = \frac{1/\sqrt{n}}{2} = \frac{1}{2\sqrt{n}}$; $f_n \rightarrow 0$ as $|x| \rightarrow \infty$.
So $M_n = \frac{1}{2\sqrt{n}} \rightarrow 0 \implies$ **uniformly to 0 on all of $\mathbb{R}$.**

TRAP: sup bounds that do and do not block uniformity

- $\sup_x |f_n(x)| \geq \frac{1}{2n}$ does **not** prevent $f_n \rightrightarrows 0$ — $\frac{1}{2n} \rightarrow 0$. Only a lower bound that *does not vanish* blocks it.
- The max being attained at a *different* point x_n for each n is **irrelevant**. Only the *value* M_n counts.
- $f_n \rightarrow f$ pointwise with all f_n and f continuous **still need not be uniform** (nxe^{-nx^2} on $[0, 1]$).

TABLE: gallery: memorise this table, it answers most (f_n) questions

f_n (domain)	ptwise f	$\ f_n - f\ _\infty$	unif.? $\int$ swaps?
x^n on $[0, 1]$	0 on $[0, 1]$, 1 at 1	1	F T
x^n on $[0, r]$, $r < 1$	0	r^n	T T
$\frac{x}{1 + nx^2}$ on $\mathbb{R}$	0	$\frac{1}{2\sqrt{n}}$	T T
$\frac{\sin(nx)}{n}$ on $\mathbb{R}$	0	$\frac{1}{n}$	T T
nxe^{-nx^2} on $[0, 1]$	0	$\sqrt{\frac{\pi}{2}}e^{-1/2}$	F F
n^2xe^{-nx} on $[0, 1]$	0	0	F F
$\arctan(nx)$ on $\mathbb{R}$	$\frac{\pi}{2}$ sgn x	$\frac{\pi}{2}$	F n/a
$\frac{x^n}{1 + x^n}$ on $[0, 1]$	0, $\frac{1}{2}$ at 1	$\frac{1}{2}$	F T

Read it as: a discontinuous limit is *never* uniform. Uniformity is sufficient but not necessary for swapping $\int$: row 1 swaps without it ($\int_0^1 x^n = \frac{1}{n+1} \rightarrow 0$), rows 5–6 do not ($\int_0^1 nxe^{-nx^2} \rightarrow \frac{1}{2}$, $\int_0^1 n^2xe^{-nx} \rightarrow 1$, both $\neq 0$).

FACT: uniform Cauchy criterion and the M -test

Uniform Cauchy: $f_n \rightrightarrows$ some limit $\iff \forall \varepsilon \exists N \forall m, n \geq N : \|f_n - f_m\|_\infty < \varepsilon$. Use it when you do not know f .
Weierstrass M -test (for $\sum f_n$): $|f_n(x)| \leq M_n \forall x \in D$ and $\sum M_n < \infty \implies \sum f_n$ converges **absolutely and uniformly** on D . This is how power series get uniform convergence on $[-r, r]$, $r < R$.
Dini: D compact, f_n continuous, f continuous, and $f_n \rightarrow f$ pointwise *monotonically* in $n \implies$ convergence is automatically uniform. The only cheap converse there is.

7.3 Interchanging limits with $\int$ and $\frac{d}{dx}$

TABLE: what you may swap, and the exact price

Swap	Hypothesis that makes it legal	Counterexample if missing
$\lim_n \int_a^b f_n = \int_a^b \lim_n f_n$ $\lim_n f'_n = (\lim_n f_n)'$	$f_n \rightrightarrows f$ on the compact $[a, b]$ $f'_n \rightrightarrows g$ uniformly and $f_n(x_0)$ converges for one x_0	n^2xe^{-nx} on $[0, 1]$: $\rightarrow 1 \neq 0$ $\frac{\sin(nx)}{n} \rightrightarrows 0$ but $f'_n = \cos(nx)$ diverges
$\sum_n \int f_n = \int \sum_n f_n$	$\sum f_n$ uniformly conv. on $[a, b]$ (M -test)	same shape as row 1
$\frac{d}{dx} \sum_n f_n = \sum_n f'_n$	$\sum f'_n$ uniformly convergent	holds free inside a power series' radius
$\lim_{x \rightarrow x_0} \lim_n f_n = \lim_n \lim_{x \rightarrow x_0} f_n$	$f_n \rightrightarrows f$ (then f is continuous)	x^n on $[0, 1]$ at $x_0 = 1$

FACT: what uniform convergence buys you

- f_n continuous on D and $f_n \rightrightarrows f$:
- f is **continuous** — the *Uniform Limit Theorem*. Pointwise alone gives nothing.
- On $[a, b]$: f is **Riemann integrable** and $\int_a^b f_n \rightarrow \int_a^b f$.
- f_n **bounded uniformly** $\implies f$ bounded.
- ! **Derivatives do not follow**. Uniform convergence of f_n says nothing about f'_n ; you must assume f'_n converges uniformly.

8	Derivatives and their applications
8.1	Differentiability
DEF: three equivalent formulations	$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0+h) - f(x_0)}{h} = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$ (two-sided, finite) $\iff \exists K \in \mathbb{R} : f(x) = f(x_0) + K(x - x_0) + o(x - x_0)$ as $x \rightarrow x_0$, and then $K = f'(x_0)$. diff. $\Rightarrow$ continuous; continuous $\Rightarrow$ diff. ($ x $ at 0).
TRAP: Landau o / O — the classic multiple-choice trap	$f = O(g) : f \leq C g $ near x_0 (same order or smaller). $f = o(g) : f/g \rightarrow 0$ (strictly smaller). $f(x) - f(x_0) - K(x - x_0) = o(x - x_0) \Rightarrow$ differentiable T — note $(x - x_0)$, not Kx: written with Kx the statement is meaningless. $f(x) - f(x_0) = O(x - x_0) \Rightarrow$ differentiable F — that is only Lipschitz near x_0; x satisfies it. - only $h \rightarrow 0^+$ exists $\Rightarrow$ differentiable F — that is only right-differentiability. “$f(x) = f(x_0) + p(x - x_0) + o(x - x_0)$ for an arbitrary polynomial p” F — useless; p must be exactly the linear Taylor part. Algebra: $o(x^n) \pm o(x^n) = o(x^n)$, $x^m \cdot o(x^n) = o(x^{m+n})$, $o(x^n) = O(x^n)$ but not conversely.
► RECIPE: piecewise function: differentiable at the seam x_0 ?	- Continuity first: $\lim_{x \rightarrow x_0^-} f = \lim_{x \rightarrow x_0^+} f = f(x_0)$. Fails $\Rightarrow$ not differentiable, stop. - Compare the one-sided difference quotients (not just $\lim f'$). Equal $\Rightarrow$ differentiable, with that value. - For C^1 additionally $\lim_{x \rightarrow x_0^-} f' = \lim_{x \rightarrow x_0^+} f'$. Ex. $f(x) = x^2 \sin \frac{1}{x}$ ($x \neq 0$), $f(0) = 0$: $f'(0) = \lim h \sin \frac{1}{h} = 0$ exists, but f' is not continuous at 0 $\Rightarrow$ differentiable, not C^1 .
8.2	Rules — and how not to misuse them
FACT: the rules	$(f \pm g)' = f' \pm g'$; $(fg)' = f'g + fg'$; $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$; chain $(g \circ f)'(x) = g'(f(x))f'(x)$; inverse $(f^{-1})'(y_0) = \frac{1}{f'(x_0)}$, $y_0 = f(x_0)$; Leibniz $(fg)^{(n)} = \sum_{k=0}^n \binom{n}{k} f^{(k)} g^{(n-k)}$; logarithmic for $u(x)^{v(x)}$: write $u^v = e^{v \ln u} \Rightarrow (u^v)' = u^v (v' \ln u + \frac{v u'}{u})$.
► RECIPE: implicit differentiation	Given $F(x, y) = 0$ with $y = y(x)$: differentiate both sides in x treating y as a function, so every y contributes a factor y' by the chain rule, then solve for y' . Ex. $x^2 + y^2 = 1 \Rightarrow 2x + 2yy' = 0 \Rightarrow y' = -\frac{x}{y}$.
TRAP: using the derivative/antiderivative table correctly	The table (§13.1) lists $F(x)$ for the bare argument x . If the argument is $ax + b$: $\int f(ax + b) dx = \frac{1}{a} F(ax + b) + C \quad (\text{divide by the inner derivative})$! Valid only for linear inner functions. Anything else: substitute properly. Worked trap: $\int \frac{1}{2 \cos^2(\frac{1}{2}x)} dx$. Table: $\int \frac{du}{\cos^2 u} = \tan u$. With $u = \frac{x}{2}$, $dx = 2du$: $\int \frac{2du}{2 \cos^2 u} = \tan u = \tan \frac{x}{2} + C$ — not $2 \tan \frac{x}{2}$. Always differentiate your answer. Five seconds, catches every chain-rule slip.
8.3	Mean value theorems (Mittelwertsätze)
FACT: Fermat / Rolle / Lagrange / Cauchy	- Fermat: x_0 an interior local extremum, f differentiable there $\Rightarrow f'(x_0) = 0$. Converse false (x^3). - Rolle: $f \in C[a, b]$, diff. on (a, b), $f(a) = f(b) \Rightarrow \exists \xi \in (a, b) : f'(\xi) = 0$. - MVT (Lagrange): same hypotheses $\Rightarrow \exists \xi \in (a, b) : f'(\xi) = \frac{f(b) - f(a)}{b - a}$. - Cauchy MVT: $\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(\xi)}{g'(\xi)}$ — this is what proves l'Hospital.
► RECIPE: the four things MVT is for	- Prove $f(x) - f(y) \leq L x - y$: MVT plus a bound $f' \leq L$. (Route to Lipschitz $\Rightarrow$ uniform continuity.) - Prove f constant: $f' \equiv 0$ on an interval $\Rightarrow$ constant. - Count zeros: n zeros of $f \Rightarrow$ at least $n - 1$ zeros of f' (Rolle between consecutive ones). Contrapositive: f' has $\leq k$ zeros $\Rightarrow f$ has $\leq k + 1$. - Compare two functions: set $g = f - h$, show $g'(a) = 0$ and $g' > 0 \Rightarrow f > h$ on (a, ∞).

► RECIPE: “show there is a c with $f'(c) = k$ ” / “with $f(c) = v$ ” — pick the sub-interval
You are given a few values of f and asked for a point with a prescribed derivative or a prescribed value. Do not try to find c ; produce it from a theorem. - Prescribed $f'(c) = k \Rightarrow$ MVT. Find the sub-interval $[p, q]$ whose secant slope already equals k: solve $\frac{f(q) - f(p)}{q - p} = k$ among the given points. Then MVT on $[p, q]$ hands you $c \in (p, q)$. Check f is continuous on $[p, q]$ and differentiable on (p, q) — say it. - Prescribed $f(c) = v \Rightarrow$ IVT. Find a sub-interval whose two endpoint values straddle v. - Prescribed $f'(c) = 0 \Rightarrow$ Rolle: find two points with equal values. Ex. f cont. on $[0, 2]$, diff. on $(0, 2)$, $f(0) = f(2) = 0$, $f(1) = 1$. $c_1 \in (0, 1)$ with $f'(c_1) = 1$: slope on $[0, 1]$ is $\frac{1 - 0}{1 - 0} = 1 \Rightarrow$ MVT. $c_2 \in (1, 2)$ with $f'(c_2) = -1$: slope on $[1, 2]$ is $\frac{0 - 1}{2 - 1} = -1 \Rightarrow$ MVT. $c_3 \in (1, 2)$ with $f(c_3) = \frac{1}{2}$: $f(1) = 1 > \frac{1}{2} > 0 = f(2) \Rightarrow$ IVT.

8.4	Curve discussion (Kurvendiskussion) and extrema
► RECIPE: find and classify all extrema of f on I	- Domain, plus every point where f is not differentiable (kinks, poles) and the endpoints of I — all are candidates. - Critical points: solve $f'(x) = 0$. - Classify: $f''(x_0) > 0 \Rightarrow$ local min; $f''(x_0) < 0 \Rightarrow$ local max; $f''(x_0) = 0 \Rightarrow$ inconclusive, use the sign change of f' ($- \rightarrow +$ min, $+ \rightarrow -$ max, no change $\Rightarrow$ saddle). - Global on $[a, b]$: evaluate f at all critical points and at a, b; largest wins (it exists, by Weierstrass). - Global on open/unbounded I: also take $\lim_{x \rightarrow aI} f$; if a limit beats every candidate, the extremum is not attained. - Monotonicity: sign table of f'. Convexity: sign table of f''; a sign change of f'' is an inflection point (Wendepunkt). - Asymptotes: $m = \lim \frac{f(x)}{x}$, $c = \lim (f(x) - mx)$; then $y = mx + c$.

EXAMPLE: full discussion of $f(x) = \frac{x}{1 + x^2}$
Domain $\mathbb{R}$, odd. $f' = \frac{1 - x^2}{(1 + x^2)^2} \Rightarrow$ critical $x = \pm 1$. $f'' = \frac{2x(x^2 - 3)}{(1 + x^2)^3} \Rightarrow f''(1) < 0$: max $f(1) = \frac{1}{2}$; $f''(-1) > 0$: min $f(-1) = -\frac{1}{2}$. Inflection at $x = 0, \pm\sqrt{3}$. Increasing on $[-1, 1]$, decreasing outside. $f \rightarrow 0$ at $\pm\infty \Rightarrow$ asymptote $y = 0$. Global max/min $= \pm\frac{1}{2}$, both attained.

FACT: convexity (konvex) — exam favourite
f convex on $I \iff f'' \geq 0 \iff f'$ increasing $\iff f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y) \iff$ the graph lies above every tangent. - convex and concave $\Rightarrow f$ is constant F — it is affine, $f(x) = ax + b$ ($f'' \equiv 0$, not $f' \equiv 0$). - Convex and $f'(x_0) = 0$ at an interior point $\Rightarrow x_0$ is a global minimum. T - A convex differentiable f can have several isolated critical points F — the critical set is always an interval. - Its global maximum on $[a, b]$ is always attained at a boundary point. T “the global minimum can only be on the boundary” F — it may well be interior.

► RECIPE: prove an inequality with convexity
- Tangent-line trick: f convex $\Rightarrow f(x) \geq f(a) + f'(a)(x - a)$ for all x. Choosing a good a proves most one-variable inequalities in one line, e.g. $e^x \geq 1 + x$ ($a = 0$), $\ln x \leq x - 1$ ($f = -\ln$, $a = 1$). - Jensen: f convex $\Rightarrow f\left(\frac{1}{n} \sum x_i\right) \leq \frac{1}{n} \sum f(x_i)$. With $f = -\ln$ this is AM–GM.

8.5	l'Hospital (Bernoulli-de l'Hospital)
► RECIPE: use it safely	- Check the form first: only $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$. Applying it unchecked loses marks. - Differentiate numerator and denominator separately — ! not the quotient rule. - Repeat while the form persists. - Verify the new limit exists. If it does not, l'Hospital states nothing about the original. Reduce other forms first: $0 \cdot \infty \Rightarrow \frac{f}{1/g}$; $\infty \cdot \infty \Rightarrow$ common denominator or conjugate; $1^\infty, 0^0, \infty^0 \Rightarrow u^v = e^{v \ln u}$, work in the exponent, exponentiate at the end.

TRAP: when l'Hospital fails
$\lim_{x \rightarrow \infty} \frac{x + \sin x}{x} = 1$, but l'Hospital gives $1 + \cos x$, which has no limit $\Rightarrow$ inapplicable. Divide by x instead. $\lim_{x \rightarrow \infty} \frac{x}{\sqrt{1 + x^2}}$: l'Hospital cycles forever. Divide by x. - Repeated use that never resolves ($\frac{e^x}{x^x}$ patterns) $\Rightarrow$ switch to Taylor or algebra. Rule of thumb: at $x \rightarrow 0$ prefer Taylor (§5.3); it is faster and cannot mislead you.

EXAMPLE: indeterminate forms, one line each
$\lim_{x \rightarrow 0^+} x \ln x = \lim \frac{\ln x}{1/x} \stackrel{\infty/\infty}{=} \lim \frac{1/x}{-1/x^2} = \lim (-x) = 0$. $\lim_{x \rightarrow 0^+} x^x = e^{\lim x \ln x} = e^0 = 1$. $\lim_{x \rightarrow \infty} \left(1 + \frac{a}{x}\right)^x = e^{\lim x \ln(1 + a/x)} = e^a$. $\lim_{x \rightarrow 0^+} \frac{\ln x}{\cot x} \cdot \frac{-\infty}{\infty} \Rightarrow$ l'Hospital $\Rightarrow \lim \frac{1/x}{-1/\sin^2 x} = -\lim \frac{\sin^2 x}{x} = 0$.

9	Integration
9.1	Which method? Match the integrand
► RECIPE: take the first match, then go to that subsection	- Already in the table (§13.1)? $\Rightarrow$ read it off. Inner function $ax + b \Rightarrow$ multiply by $\frac{1}{a}$, nothing else may be inner. - The inner derivative appears as a factor, $\int g(f(x))f'(x) dx \Rightarrow$ substitution $u = f(x)$, §9.2. - Product of two unrelated types (poly $\times e^x/\sin/\ln/\arctan$) $\Rightarrow$ by parts, §9.3. - Rational function $\frac{p}{q} \Rightarrow$ division if $\deg p \geq \deg q$, then partial fractions, §9.4. - Rational in $\sin, \cos \Rightarrow$ half-angle identities first, else $t = \tan \frac{x}{2}$, §9.5. $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$, $\sqrt{x^2 - a^2} \Rightarrow$ trig substitution $x = a \sin \theta / a \tan \theta / a \sec \theta$. $\sqrt[n]{ax + b} \Rightarrow$ substitute $u = \sqrt[n]{ax + b}$; the root disappears completely. $\sin^n x \cos^n x \Rightarrow$ one odd power: peel a factor off and substitute the other. Both even: half-angle. $\sin^n x$, $\cos^n x$, $x^n e^x$ with large $n \Rightarrow$ reduction formula, §9.3. - Nothing works $\Rightarrow$ symmetry (§9.6), or the answer legitimately contains $\int \frac{\sin x}{x} dx$, which is non-elementary and allowed in a multiple-choice option. Always differentiate your result at the end. Under time pressure it is the only check that matters.

9.2	Substitution
► RECIPE: step by step	- Choose $u = g(x)$, usually the inner function of a composition. $du = g'(x) dx \Rightarrow$ solve for dx and make every x disappear. - Definite: convert the bounds $a \mapsto g(a)$, $b \mapsto g(b)$ — then do not substitute back. - Indefinite: integrate in u, then substitute $u = g(x)$ back. ! Differentiate the result: the chain-rule factor $g'(x)$ must cancel. This is where most marks are lost. Reverse direction ($x = \varphi(t)$, φ bijective and C^1): $\int f(x) dx = \int f(\varphi(t))\varphi'(t) dt$, back via $t = \varphi^{-1}(x)$.

FACT: the three substitution patterns worth memorising
$\int \frac{f'}{f} dx = \ln f + C$; $\int f^n f' dx = \frac{f^{n+1}}{n+1} + C$; $\int f' e^f dx = e^f + C$; $\int \frac{f'}{1 + f^2} dx = \arctan f + C$. Ex. $\int \frac{x}{1 + x^2} dx$: $u = 1 + x^2$, $du = 2x dx \Rightarrow \frac{1}{2} \int \frac{du}{u} = \frac{1}{2} \ln(1 + x^2) + C$.

EXAMPLE: trig substitution done properly
$\int \sqrt{1 - x^2} dx$: put $x = \sin \theta$, $dx = \cos \theta d\theta \Rightarrow \int \cos^2 \theta d\theta = \frac{\theta}{2} + \frac{\sin 2\theta}{4}$. Back-substitute with $\theta = \arcsin x$ and $\sin 2\theta = 2 \sin \theta \cos \theta = 2x\sqrt{1 - x^2}$: $\int \sqrt{1 - x^2} dx = \frac{1}{2} \arcsin x + \frac{1}{2} x\sqrt{1 - x^2} + C$! Back-substitution needs the identity, not just $\theta = \arcsin x$.

9.3	Integration by parts (partielle Integration)
► RECIPE: $\int uv' dx = uv - \int u'v dx$	- Choose u = the factor that gets simpler when differentiated. Priority LIATE: Log, Inverse trig, Algebraic, Trig, Exponential — earlier in the list becomes u. v' = the rest; integrate it to get v (no $+C$ here). - Apply, repeat if the new integral is still a product ($x^2 e^x$: twice). $\int \ln x dx$, $\int \arctan x dx$: take $v' = 1 \Rightarrow x \ln x - x$, $x \arctan x - \frac{1}{2} \ln(1 + x^2)$.
► RECIPE: cycle trick (when the integral comes back)	$I = \int e^x \sin x dx$: two rounds of parts return $I = -e^x \cos x + e^x \sin x - I$. Treat I as an unknown and solve algebraically: $2I = e^x (\sin x - \cos x) \Rightarrow I = \frac{e^x (\sin x - \cos x)}{2} + C$. Same for $\int e^{ax} \cos(bx) dx$ and $\int \sec^3 x dx$.

FACT: reduction formulas

$$\int \sin^n x \, dx = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} \int \sin^{n-2} x \, dx$$
$$\int \cos^n x \, dx = \frac{\cos^{n-1} x \sin x}{n} + \frac{n-1}{n} \int \cos^{n-2} x \, dx$$
$$\int x^n e^x \, dx = x^n e^x - n \int x^{n-1} e^x \, dx; \quad \int (\ln x)^n \, dx = x (\ln x)^n - n \int (\ln x)^{n-1} \, dx$$

Iterate down to $n = 0$ or 1 . Definite version on $[0, \frac{\pi}{2}]$: $\int_0^{\pi/2} \sin^n = \frac{n-1}{n} \int_0^{\pi/2} \sin^{n-2}$.

EXAMPLE: multiple choice: “ $\int x \ln x \sin x \, dx = \dots$ ”

Do **not** integrate. **Differentiate the four options** and compare with the integrand: 20 seconds instead of 5 minutes.
Check of $-x \ln x \cos x + \ln x \sin x + \sin x - \int \frac{\sin x}{x} \, dx$:
 $-(\ln x + 1) \cos x + x \ln x \sin x + \frac{\sin x}{x} + \ln x \cos x + \cos x - \frac{\sin x}{x} = x \ln x \sin x \quad \checkmark$

9.4 Partial fractions (Partialbruchzerlegung)

► RECIPE: $\int \frac{p(x)}{q(x)} \, dx$

- deg $p \geq \deg q \Rightarrow$ polynomial division first.
- Factor q completely over $\mathbb{R}$: linear factors and irreducible quadratics only (§2.3).
- Ansatz: $(x-a) \rightarrow \frac{A}{x-a}$; $(x-a)^k \rightarrow \frac{A_1}{x-a} + \dots + \frac{A_k}{(x-a)^k}$; irreducible $x^2+bx+c \rightarrow \frac{Ax+B}{x^2+bx+c}$.
- Multiply out and compare coefficients, or use the **cover-up rule** — fastest for simple linear factors: to get the A belonging to $\frac{A}{x-a}$, **cover up** the factor $(x-a)$ in q and evaluate what is left at $x=a$, i.e. $A = \frac{p(a)}{q'(a)}$. Ex. $\frac{1}{x^2-1} = \frac{1}{x-1} + \frac{B}{x+1}$: $A = \frac{1}{1-1} = \frac{1}{2}$, $B = \frac{1}{-1-1} = -\frac{1}{2}$.
- Integrate: $\int \frac{A}{x-a} = A \ln|x-a|$; $\int \frac{A}{(x-a)^k} = \frac{A}{(1-k)(x-a)^{k-1}}$; for $\frac{Ax+B}{x^2+bx+c}$ split off $\frac{2x+b}{x^2+bx+c}$ (gives $\ln$) and complete the square on the rest (gives arctan).

Ex. $\int \frac{dx}{x^2-1} = \int \left(\frac{1/2}{x-1} - \frac{1/2}{x+1} \right) dx = \frac{1}{2} \ln \left| \frac{x-1}{x+1} \right| + C$.

9.5 Rational in sin and cos

► RECIPE: half-angle first, Weierstrass only if needed

Try first: $1 + \cos x = 2 \cos^2 \frac{x}{2}$, $1 - \cos x = 2 \sin^2 \frac{x}{2}$, $\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$.
Weierstrass $t = \tan \frac{x}{2}$: $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$, $dx = \frac{2}{1+t^2} dt \Rightarrow$ rational in $t \Rightarrow$ §9.4.

EXAMPLE: $\int \frac{dx}{1+\cos x}$ (and its twin)

$1 + \cos x = 2 \cos^2 \frac{x}{2} \Rightarrow \int \frac{dx}{2 \cos^2 \frac{x}{2}} \stackrel{u=x/2}{=} \int \frac{2 du}{2 \cos^2 u} = \tan u = \boxed{\tan \frac{x}{2} + C}$.

Check: $\frac{d}{dx} \tan \frac{x}{2} = \frac{1}{2 \cos^2 \frac{x}{2}} = \frac{1}{1+\cos x} \quad \checkmark$
Twin: $\int \frac{dx}{1-\cos x} = -\cot \frac{x}{2} + C$.

9.6 Definite integrals, FTC, symmetry

FACT: Fundamental theorem (Hauptsatz)

f continuous on $[a, b]$, $F(x) := \int_a^x f(t) \, dt \Rightarrow F$ is differentiable and $F' = f$.
Newton–Leibniz: $\int_a^b f = F(b) - F(a)$ for any antiderivative F .
Variable bounds (chain rule!): $\frac{d}{dx} \int_{u(x)}^{v(x)} f(t) \, dt = f(v(x))v'(x) - f(u(x))u'(x)$.
MVT for integrals: f continuous $\Rightarrow \exists \xi \in [a, b]$ with $\int_a^b f = f(\xi)(b-a)$.

FACT: free simplifications

- f odd on a symmetric interval $\Rightarrow \int_{-a}^a f = 0$; f even $\Rightarrow = 2 \int_0^a f$.
- $\int_a^b = -\int_b^a$; $\int_a^b = \int_a^c + \int_c^b$; $|\int f| \leq \int |f|$; $f \leq g \Rightarrow \int f \leq \int g$.
- $\left| \int_a^b f \right| \leq (b-a) \max |f|$ (the standard estimate).
- Period T : $\int_a^{a+T} f$ does not depend on a .
- $\int_0^{2\pi} \sin^2 = \int_0^{2\pi} \cos^2 = \pi$; $\int_0^{\pi} \sin x \, dx = 2$; $\int_0^{\pi/2} \sin^2 = \frac{\pi}{4}$.

► RECIPE: limit of a sum = integral (Riemann sum trick)

A limit of the shape $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right)$ equals $\int_0^1 f(x) \, dx$.

- Factor out $\frac{1}{n}$ (the width). If it is not there, the trick does not apply.
- Write the rest as a function of $\frac{k}{n}$ alone $\Rightarrow$ that is f .
- General interval: $\frac{b-a}{n} \sum f\left(a + k \frac{b-a}{n}\right) \rightarrow \int_a^b f$.

Ex. $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n}{n^2+k^2} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum \frac{1}{1+(k/n)^2} = \int_0^1 \frac{dx}{1+x^2} = \frac{\pi}{4}$.
Ex. $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \ln \frac{k}{n} = \int_0^1 \ln x \, dx = -1$.

► RECIPE: integrals with $|\cdot|$, max, or pieces

- Find every point where the sign or the active branch changes (zeros of the inside).
- Split there, drop the $|\cdot|$ with the correct sign on each piece.
- Add up. !Never pull $|\cdot|$ outside the integral.

Ex. $\int_0^2 |x-1| \, dx = \int_0^1 (1-x) \, dx + \int_1^2 (x-1) \, dx = \frac{1}{2} + \frac{1}{2} = 1$.

9.7 Riemann integrability (Riemann-integrierbar)

DEF: Darboux definition

Partition $P: a = x_0 < \dots < x_n = b$; $L(f, P) = \sum_i (\inf_{[x_{i-1}, x_i]} f) \Delta x_i$, $U(f, P) = \sum_i (\sup_{[x_{i-1}, x_i]} f) \Delta x_i$.
 f (bounded) is **integrable** $\iff \sup_P L = \inf_P U \iff \forall \epsilon > 0 \exists P: U(f, P) - L(f, P) < \epsilon$.
To compute from the definition: take the equidistant P , evaluate the two sums, let $n \rightarrow \infty$.

FACT: criteria — and the false friends

- continuous on $[a, b] \Rightarrow$ integrable. **T** Monotone on $[a, b] \Rightarrow$ integrable. Differentiable $\Rightarrow$ continuous $\Rightarrow$ integrable.
- bounded with *finitely* (even countably) many discontinuities $\Rightarrow$ integrable. **T** Lebesgue criterion: bounded and the discontinuity set is a null set $\iff$ integrable.
- “must be continuous everywhere” **F** — jumps are fine.
- “integrable $\Rightarrow$ differentiable” **F** — a step function is integrable and not even continuous.
- “finitely many function values $\Rightarrow$ integrable” **F** — Dirichlet 1_Q takes only 0, 1 and is not integrable.
- ! Riemann integrability requires **bounded** f ; unbounded means improper.

9.8 Improper integrals (uneigentliche Integrale)

► RECIPE: does it converge?

- Locate every problem point: $\pm\infty$ and every point where f is unbounded. !Split so each piece has *exactly* one problem; all pieces must converge.
- Write each as a limit: $\int_a^\infty f = \lim_{R \rightarrow \infty} \int_a^R f$, $\int_a^b f = \lim_{\epsilon \rightarrow 0^+} \int_a^{b-\epsilon} f$.
- Compare with the benchmark below at that specific point.
- Limit comparison: $\frac{f(x)}{g(x)} \rightarrow c \in (0, \infty) \Rightarrow$ same behaviour; replace f by its leading behaviour there.
- Sign changes: test absolute convergence first; if that fails, oscillating integrals may still converge conditionally (parts / Dirichlet).
- !An interior singularity does **not** cancel by symmetry: $\int_{-1}^1 \frac{dx}{x}$ **diverges**, because $\int_0^1 \frac{dx}{x}$ already does. Both halves must converge *separately* — there is no principal value in this course.

TABLE: benchmarks: the exponent flips meaning at the two ends

Integral	Converges iff
$\int_1^\infty x^{-p} \, dx$	$p > 1$ (needs <i>fast</i> decay at ∞)
$\int_0^1 x^{-p} \, dx$	$p < 1$ (only a <i>mild</i> singularity at 0)
$\int_0^\infty \frac{dx}{x^p}$	$p > 1$
$\int_0^1 \frac{1}{x^p} \frac{dx}{x^p}$	$p > 1$
$\int_0^\infty e^{-\lambda x} \, dx = \frac{1}{\lambda}$	$\lambda > 0$
$\int_1^\infty \frac{\sin x}{x} \, dx$	converges, <i>conditionally</i> (parts)
$\int_0^1 \ln x \, dx = -1$	converges

EXAMPLE: inspect both ends before answering

- $\int_0^\infty \frac{\sin t}{t} \, dt$: at $t \rightarrow 0$, $\frac{\sin t}{t} \sim \frac{1}{t} \Rightarrow$ **diverges**. The problem is at 0, not at ∞ .
- $\int_0^\infty \frac{dt}{t^2+t^4}$: at $t \rightarrow 0$, $\sim \frac{1}{t^2} \Rightarrow$ **diverges**.
- $\int_1^\infty \frac{\cos t}{t} \, dt$: $\cos^2 t = \frac{1+\cos 2t}{2}$ contains $\int \frac{dt}{2t} \Rightarrow$ **diverges**.
- $\int_1^\infty \frac{\cos t}{t} \, dt$: parts with $u = \frac{1}{t} \Rightarrow$ **converges** conditionally.

10 Differential equations (ODE)

10.1 Which method? Classify first

► RECIPE: read the shape, then jump

- $y' = f(x)g(y)$ (factorises) $\Rightarrow$ **separation**, §10.2.
- $y' + p(x)y = q(x)$ (linear, 1st order) $\Rightarrow$ **integrating factor**, §10.3.
- $y' + p(x)y = q(x)y^\alpha \Rightarrow$ **Bernoulli**, $z = y^{1-\alpha}$, §10.4.
- $y' = f(\frac{y}{x}) \Rightarrow u = \frac{y}{x}$, i.e. $y = ux$, $y' = u'x + u \Rightarrow$ separable.
- $y' = f(ax+by+c) \Rightarrow u = ax+by+c \Rightarrow$ separable.
- $y^{(n)} + a_{n-1}y^{(n-1)} + \dots + a_0y' = g(x)$, constant $a_i \Rightarrow$ characteristic polynomial (§10.5) + particular solution (§10.6).
- $y' = f(y)$ (**autonomous**) $\Rightarrow$ separable; the constant solutions are the zeros of f , and they decide the qualitative picture (§10.7).

Order of work, always: general solution first $\Rightarrow$ then insert the initial/boundary conditions $\Rightarrow$ then solve for the constants.

10.2 Separation of variables (Trennung der Variablen)

► RECIPE: $y' = f(x)g(y)$

- ! First solve $g(y_0) = 0$: each root gives a **constant solution** $y \equiv y_0$. Forgetting these costs marks and they are often the ones the initial condition needs.
- For $g(y) \neq 0$: $\frac{dy}{g(y)} = f(x) \, dx$, integrate both sides, one constant C on the right.
- Solve for y ; handle $|\cdot|$ and $\pm$ by absorbing into C ($e^C =: \tilde{C} > 0$, the sign extends it to $\tilde{C} \in \mathbb{R}$).
- Insert the initial condition, then state the interval of existence.

Ex. $y' = xy$: $\frac{dy}{y} = x \, dx \Rightarrow \ln|y| = \frac{x^2}{2} + C \Rightarrow y = \tilde{C}e^{x^2/2}$ (which includes $y \equiv 0$ at $\tilde{C} = 0$).

EXAMPLE: separable with an initial condition

$y' = y^2 \cos x$, $y(0) = 1$. Constant solution $y \equiv 0$ (does not meet $y(0) = 1$).

Otherwise $-\frac{1}{y} = \sin x + C \Rightarrow y = \frac{-1}{\sin x + C}$; $y(0) = 1 \Rightarrow C = -1 \Rightarrow \boxed{y = \frac{1}{1 - \sin x}}$, valid near $x = 0$ until $\sin x = 1$.

10.3 First order linear $y' + p(x)y = q(x)$

► RECIPE: integrating factor (fastest, always works)

- Normal form $y' + p(x)y = q(x)$ — the coefficient of y' must be 1.
- $\mu(x) := e^{\int p(x) \, dx}$ (no $+C$ needed).
- $(\mu y)' = \mu q \Rightarrow y = \frac{1}{\mu(x)} \left(\int \mu(x)q(x) \, dx + C \right)$.
- Insert the initial condition last.

“Variation of constants” (*Variation der Konstanten*) is the same method, spelled out: solve the homogeneous problem first, $y_h = Ce^{-\int p}$; then **promote the constant to a function** $C = C(x)$ and insert $y = C(x)e^{-\int p}$ into the ODE. Everything cancels except $C' = qe^{\int p}$, which you integrate. Use the name if the exam asks for it; the integrating factor gets you there faster.

EXAMPLE: $y' - \frac{2}{x}y = x^3$

$p = -\frac{2}{x} \Rightarrow \mu = e^{-2 \ln|x|} = x^{-2}$. Then $(x^{-2}y)' = x \Rightarrow x^{-2}y = \frac{x^2}{2} + C \Rightarrow \boxed{y = \frac{x^4}{2} + Cx^2}$.

10.4 Bernoulli and substitutions

► RECIPE: $y' + p(x)y = q(x)y^\alpha$, $\alpha \neq 0, 1$

- Divide by y^α : $y^{-\alpha}y' + py^{1-\alpha} = q$.
- Set $z := y^{1-\alpha} \Rightarrow z' = (1-\alpha)y^{-\alpha}y'$.
- $\Rightarrow$ **linear** ODE $z' + (1-\alpha)pz = (1-\alpha)q \Rightarrow$ §10.3.
- Back-substitute $y = z^{1/(1-\alpha)}$; check the lost solution $y \equiv 0$ separately.

Ex. $y' + y = xy^2$ ($\alpha = 2$): $z = y^{-1} \Rightarrow z' - z = -x \Rightarrow \mu = e^{-x} \Rightarrow z = x + 1 + Ce^x \Rightarrow y = \frac{1}{x+1+Ce^x}$.

10.5 Constant coefficients: homogeneous part

► RECIPE: $y^{(n)} + a_{n-1}y^{(n-1)} + \dots + a_0y = 0$

- Ansatz $y = e^{\lambda x} \Rightarrow$ **characteristic polynomial** $\lambda^n + a_{n-1}\lambda^{n-1} + \dots + a_0 = 0$.
- Find all roots** (guess an integer root, then polynomial division).
- Build the fundamental system:
 - simple real root $\lambda \Rightarrow e^{\lambda x}$
 - real root of multiplicity $m \Rightarrow e^{\lambda x}, x e^{\lambda x}, \dots, x^{m-1} e^{\lambda x}$
 - complex pair $\alpha \pm i\beta \Rightarrow e^{\alpha x} \cos \beta x, e^{\alpha x} \sin \beta x$ (multiplicity m : also times $x, \dots, x^{m-1}$)
- $y_h = C_1 y_1 + \dots + C_n y_n$ — exactly n free constants, no more, no fewer.

2nd order quick table ($y'' + by' + cy = 0$, $D = b^2 - 4c$): $D > 0 \Rightarrow C_1 e^{\lambda_1 x} + C_2 e^{\lambda_2 x}$; $D = 0 \Rightarrow (C_1 + C_2 x) e^{\lambda x}$; $D < 0 \Rightarrow e^{\alpha x} (C_1 \cos \beta x + C_2 \sin \beta x)$.

FACT: Wronskian: is my fundamental system valid?

$W(x) = \det \begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} = y_1 y_2' - y_1' y_2$.
 $W(x_0) \neq 0$ for one $x_0 \Rightarrow y_1, y_2$ are **linearly independent** $\Rightarrow$ they span all solutions. For $y'' + by' + cy = 0$: $W(x) = W(0)e^{-bx}$, so W is either never zero or identically zero (Abel).

FACT: reduction of order (one solution known)

Given one solution y_1 of $y'' + py' + qy = 0$, set $y_2 = v(x)y_1$. Inserting kills the v term and leaves a **first-order ODE** for v' : $v' = \frac{1}{y_1} e^{-\int p}$, so $y_2 = y_1 \int \frac{e^{-\int p}}{y_1^2} \, dx$.

10.6 Particular solution: ansatz by the right-hand side

TABLE: ansatz table — m is the resonance multiplicity

$g(x)$	ansatz y_p
polynomial, degree k	$x^m (A_k x^k + \dots + A_0)$, full polynomial
$Ce^{\mu x}$	$x^m A e^{\mu x}$
$C \cos \omega x$ or $C \sin \omega x$	$x^m (A \cos \omega x + B \sin \omega x)$, both terms
$P_k(x)e^{\mu x}$	$x^m e^{\mu x} (A_k x^k + \dots + A_0)$
$e^{\mu x} \cos \omega x$	$x^m e^{\mu x} (A \cos \omega x + B \sin \omega x)$
$g_1 + g_2$	solve separately and add (superposition)
m = multiplicity of μ (resp. of $\mu \pm i\omega$) as a root of the characteristic polynomial. No resonance $\Rightarrow m = 0 \Rightarrow$ no extra x .	

► RECIPE: execute the ansatz

1. Pick the shape from the table above.

2. ! RESONANCE CHECK: is the exponent μ (resp. $\pm i\omega$) a root of the characteristic polynomial, and with what multiplicity m ? $\Rightarrow$ multiply the whole ansatz by x^m . Skipping this makes the ansatz collapse to $0 = g$.

3. Differentiate the ansatz, insert into the ODE, compare coefficients, solve for $A, B, \dots$

4. General solution = $y_h + y_p$; only now impose the initial conditions, on the *sum*.

EXAMPLE: $y'' - y = e^x$ (textbook resonance)

1) $\lambda^2 - 1 = 0 \Rightarrow \lambda = \pm 1 \Rightarrow y_h = C_1 e^x + C_2 e^{-x}$.

2) $\mu = 1$ is a simple root $\Rightarrow$ resonance, $m = 1 \Rightarrow$ ansatz $y_p = A x e^x$.

3) $y_p' = A(1+x)e^x$, $y_p'' = A(2+x)e^x \Rightarrow y_p'' - y_p = 2Ae^x \stackrel{!}{=} e^x \Rightarrow A = \frac{1}{2}$.

4) $y = C_1 e^x + C_2 e^{-x} + \frac{1}{2} x e^x$

EXAMPLE: resonant $\cos$, and a full 2nd-order IVP

(a) $y'' + y = \cos x$: $\lambda = \pm i \Rightarrow y_h = C_1 \cos x + C_2 \sin x$; $\pm i$ is a root $\Rightarrow m = 1 \Rightarrow y_p = x(A \cos x + B \sin x) \Rightarrow A = 0, B = \frac{1}{2} \Rightarrow y = C_1 \cos x + C_2 \sin x + \frac{x}{2} \sin x$.

(b) IVP $y'' - 3y' + 2y = 0$, $y(0) = 1$, $y'(0) = 0$: $\lambda = 1, 2 \Rightarrow y = C_1 e^x + C_2 e^{2x}$. $C_1 + C_2 = 1$, $C_1 + 2C_2 = 0 \Rightarrow C_2 = -1, C_1 = 2 \Rightarrow y = 2e^x - e^{2x}$.

FACT: variation of constants (when no ansatz fits, e.g. $g = \frac{1}{x} \cdot \tan x$)

2nd order with $y_h = C_1 y_1 + C_2 y_2$: set $y_p = c_1(x) y_1 + c_2(x) y_2$ with

$$c_1' = -\frac{y_2 g}{W}, \quad c_2' = \frac{y_1 g}{W}, \quad W = y_1 y_2' - y_1' y_2,$$

then integrate. **Always works**; just slower than the ansatz.

10.7 Conditions, existence, qualitative behaviour

FACT: IVP vs BVP, Picard–Lindelöf

IVP (Anfangswertproblem): all conditions at one point, $y(x_0), y'(x_0), \dots$ — n of them for order n .

BVP (Randwertproblem): conditions at two points a, b — may have 0, exactly 1, or infinitely many solutions.

Picard–Lindelöf: $y' = f(x, y)$, $y(x_0) = y_0$ with f continuous and Lipschitz in y (e.g. $\partial_y f$ continuous and locally bounded) $\Rightarrow$ a **unique** solution on a neighbourhood of x_0 . Higher order: same after rewriting as a system.

! Continuity of f alone gives existence (Peano) but **not** uniqueness: $y' = \sqrt{|y|}$, $y(0) = 0$ has both $y \equiv 0$ and $y = \frac{x^2}{4}$.

Linear ODEs with continuous coefficients: unique solution on the *whole* interval.

► RECIPE: autonomous $y' = f(y)$: sketch without solving

1. Equilibria: the roots of $f(y) = 0$ — these are the constant solutions.

2. Sign of f between consecutive equilibria: $f > 0 \Rightarrow$ solutions increase, $f < 0 \Rightarrow$ decrease.

3. An equilibrium y^* is **stable** if f goes $\rightarrow -$ there (equivalently $f'(y^*) < 0$), **unstable** if $\rightarrow +$ ($f'(y^*) > 0$).

4. Solutions can **never cross an equilibrium** (uniqueness), so each one is trapped between two of them and converges monotonically.

11 Standard proofs from the lecture (Vorlesungsbeweise)

11.1 How to write a proof for marks

► RECIPE: generic scheme

1. Write the hypotheses and the goal (“Voraussetzung / Zu zeigen”).

2. **Reduce w.l.o.g.** (oBdA) — e.g. replace f by $-f$ to fix a sign.

3. Build the auxiliary object: a set, a helper function, a subsequence. *Most marks sit here*.

4. Justify that it exists (**completeness** / **Bolzano–Weierstrass** / **Weierstrass max**).

5. Run the ε -argument or the **contradiction**.

6. Close with the statement and $\square$.

Cheap marks: name every theorem you use and quote its hypotheses (“ f continuous on the compact $[a, b]$, hence ...”). Writing out the definitions of the given and the wanted object is often already half the marks.

11.2 Tier 1 — expect one of these

FACT: intermediate value theorem (Zwischenwertsatz) (9 P)

$f \in C([a, b])$, c between $f(a)$ and $f(b) \Rightarrow \exists x^* : f(x^*) = c$.

1. oBdA $f(a) \leq c \leq f(b)$ (else consider $-f$).

2. $X := \{x \in [a, b] \mid f(x) \leq c\}$.

3. $X \neq \emptyset$ ($a \in X$) and $X \subseteq [a, b]$ is **bounded** $\Rightarrow x^* := \sup X$ exists by **completeness**.

4. **Choose** $x_n \in X \cap [x^* - \frac{1}{n}, x^*] \Rightarrow x_n \rightarrow x^*$ with $f(x_n) \leq c$.

5. **Sequential continuity** $\Rightarrow f(x^*) = \lim f(x_n) \leq c$.

6. **Suppose** $f(x^*) < c$. Then $x^* < b$ (as $c \leq f(b)$). With $\varepsilon := c - f(x^*) > 0$ **continuity** gives $\delta > 0$ such that $|y - x^*| < \delta \Rightarrow |f(y) - f(x^*)| < \varepsilon$.

7. Hence $f(y) < f(x^*) + \varepsilon = c$ there, i.e. $(x^*, x^* + \delta) \cap [a, b] \subseteq X$.

8. That **contradicts** $x^* = \sup X \Rightarrow f(x^*) = c$. $\square$

FACT: Cauchy sequence converges (Vollständigkeit von $\mathbb{R}$) (6 P)

1. **Cauchy** $\Rightarrow$ **bounded**: with $\varepsilon = 1$ get N , then $|a_n| \leq \max\{|a_1|, \dots, |a_N|, |a_N| + 1\}$.

2. **Bolzano–Weierstrass** $\Rightarrow$ a convergent subsequence $a_{n_k} \rightarrow L$.

3. **Cauchy**: $\exists N$ with $|a_n - a_m| < \frac{\varepsilon}{2}$ for all $n, m \geq N$.

4. **Subsequence**: $\exists N_1$ with $|a_{n_k} - L| < \frac{\varepsilon}{2}$ for all $k \geq N_1$.

5. $N^* := \max\{N, N_1\}$; since $n_k \geq k$, pick one index $n_k \geq N^*$.

6. **Triangle inequality**: $|a_n - L| \leq |a_n - a_{n_k}| + |a_{n_k} - L| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$ for all $n \geq N^*$. $\square$

FACT: Mean value theorem (Mittelwertsatz) (3 P)

1. **Auxiliary function** $g(x) := f(x) - \frac{f(b) - f(a)}{b - a} (x - a)$.

2. $g \in C[a, b]$, **differentiable** on (a, b) , and $g(a) = f(a) = g(b)$ — the hypotheses of **Rolle**.

3. **Rolle** $\Rightarrow \exists \xi \in (a, b)$ with $0 = g'(\xi) = f'(\xi) - \frac{f(b) - f(a)}{b - a}$. $\square$

FACT: Rolle

1. **f continuous** on the **compact** $[a, b] \Rightarrow$ attains max and min (**Weierstrass**).

2. If both are attained at the endpoints, f is constant $\Rightarrow f' \equiv 0$, done.

3. Otherwise an extremum is interior, at some $\xi \Rightarrow$ **Fermat** $\Rightarrow f'(\xi) = 0$. $\square$

FACT: Bolzano–Weierstrass

Bisection. $(a_n) \subseteq [A, B]$. Halve the interval and keep a half containing **infinitely many** terms; repeat $\Rightarrow$ nested intervals of length $\frac{B-A}{2^k}$. **Pick** indices $n_1 < n_2 < \dots$ with a_{n_k} in the k -th interval. The **nested-interval principle** gives a single point L , and $|a_{n_k} - L| \leq \frac{B-A}{2^k} \rightarrow 0$. $\square$

FACT: monotone + bounded $\Rightarrow$ convergent

(a_n) **increasing** and **bounded** $\Rightarrow s := \sup\{a_n\}$ exists (**completeness**). Given $\varepsilon > 0$, $s - \varepsilon$ is not an upper bound $\Rightarrow \exists N : a_N > s - \varepsilon$. **Monotonicity** $\Rightarrow s - \varepsilon < a_N \leq a_n \leq s$ for all $n \geq N \Rightarrow |a_n - s| < \varepsilon$. $\square$

11.3 Tier 2 — likely

FACT: extreme value theorem (Weierstrass)

(i) **bounded**: otherwise $\exists x_n$ with $|f(x_n)| \geq n$; **Bolzano–Weierstrass** $\Rightarrow x_{n_k} \rightarrow x^* \in [a, b]$, **continuity** $\Rightarrow f(x_{n_k}) \rightarrow f(x^*)$ finite — **contradiction**.

(ii) **attained**: $S := \sup f$ exists by (i); pick x_n with $f(x_n) \rightarrow S$; a convergent subsequence $x_{n_k} \rightarrow x^*$ gives $f(x^*) = S$ by **continuity**. $\square$

FACT: continuous on a compact interval $\Rightarrow$ uniformly continuous (Heine)

Suppose not: $\exists \varepsilon_0 > 0$ and x_n, y_n with $|x_n - y_n| < \frac{1}{n}$ but $|f(x_n) - f(y_n)| \geq \varepsilon_0$. **Bolzano–Weierstrass** $\Rightarrow x_{n_k} \rightarrow x^*$, and then $y_{n_k} \rightarrow x^*$ too. **Continuity** $\Rightarrow |f(x_{n_k}) - f(y_{n_k})| \rightarrow 0$ — **contradiction**. $\square$

FACT: fundamental theorem of calculus

$F(x) = \int_a^x f$. Then $\frac{F(x+h) - F(x)}{h} = \frac{1}{h} \int_x^{x+h} f = f(\xi_h)$ for some ξ_h between x and $x + h$ (MVT for integrals). As $h \rightarrow 0$, $\xi_h \rightarrow x$, so **continuity** gives $F'(x) = f(x)$. $\square$

FACT: uniform limit of continuous functions is continuous

The $\frac{\varepsilon}{3}$ -argument: $|f(x) - f(x_0)| \leq |f(x) - f_N(x)| + |f_N(x) - f_N(x_0)| + |f_N(x_0) - f(x_0)|$. **Choose** N with $\|f_N - f\|_\infty < \frac{\varepsilon}{3}$ (**uniformity**: the same N for all x — this is the whole point), then δ from continuity of f_N at x_0 . $\square$

FACT: absolute convergence $\Rightarrow$ convergence

Cauchy criterion for series: $|\sum_{k=m}^m a_k| \leq \sum_{k=m}^m |a_k| < \varepsilon$ for $m \geq n \geq N$. So the partial sums are Cauchy, hence convergent. $\square$

FACT: Fermat

x_0 an interior local max: for $h > 0$ small, $\frac{f(x_0+h) - f(x_0)}{h} \leq 0$; for $h < 0$ it is ≥ 0 . Both one-sided limits equal $f'(x_0)$, so $f'(x_0) = 0$. $\square$

FACT: differentiable $\Rightarrow$ continuous

$f(x) - f(x_0) = \frac{f(x) - f(x_0)}{x - x_0} \cdot (x - x_0) \rightarrow f'(x_0) \cdot 0 = 0$. $\square$

11.4 Tier 3 — one-liners you should still know

FACT: convergent $\Rightarrow$ bounded

$\varepsilon = 1$ gives N with $|a_n - a| < 1$ for $n \geq N \Rightarrow |a_n| \leq |a| + 1$; the finitely many earlier terms are bounded by their maximum. $\square$

FACT: convergent $\Rightarrow$ Cauchy

$|a_n - a_m| \leq |a_n - a| + |a - a_m| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2}$ for $n, m \geq N$. $\square$

FACT: the limit is unique

Assume $a_n \rightarrow a$ and $a_n \rightarrow b$ with $a \neq b$. Take $\varepsilon = \frac{|a-b|}{2}$: then $|a - b| \leq |a - a_n| + |a_n - b| < |a - b|$ — **contradiction**. $\square$

FACT: squeeze theorem

$b_n \leq a_n \leq c_n$ with $b_n, c_n \rightarrow L$: for large n , $L - \varepsilon < b_n \leq a_n \leq c_n < L + \varepsilon$. $\square$

FACT: limit laws (sum, product)

Sum: $|(a_n + b_n) - (a + b)| \leq |a_n - a| + |b_n - b| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2}$.

Product: $|a_n b_n - ab| \leq |a_n| |b_n - b| + |b| |a_n - a|$, and (a_n) is bounded by some C (convergent $\Rightarrow$ bounded) $\Rightarrow \leq C |b_n - b| + |b| |a_n - a| < \varepsilon$. $\square$

FACT: product rule

$\frac{(fg)(x) - (fg)(x_0)}{x - x_0} = f(x) \frac{g(x) - g(x_0)}{x - x_0} + g(x_0) \frac{f(x) - f(x_0)}{x - x_0}$; let $x \rightarrow x_0$ and use $f(x) \rightarrow f(x_0)$. $\square$

FACT: chain rule

Set $\varphi(y) := \frac{g(y) - g(y_0)}{y - y_0}$ for $y \neq y_0$ and $\varphi(y_0) := g'(y_0)$; then φ is continuous at y_0 and $g(y) - g(y_0) = \varphi(y)(y - y_0)$ for all y . Insert $y = f(x)$, $y_0 = f(x_0)$, divide by $x - x_0$, let $x \rightarrow x_0$. $\square$

FACT: derivative of the inverse

f^{-1} continuous and $f'(x_0) \neq 0$: $\frac{f^{-1}(y) - f^{-1}(y_0)}{y - y_0} = \frac{x - x_0}{f(x) - f(x_0)} \rightarrow \frac{1}{f'(x_0)}$ as $y \rightarrow y_0$ (which forces $x \rightarrow x_0$). $\square$

FACT: geometric series

$s_N = \sum_{k=0}^N q^k$; then $s_N - q s_N = 1 - q^{N+1} \Rightarrow s_N = \frac{1 - q^{N+1}}{1 - q}$. For $|q| < 1$, $q^{N+1} \rightarrow 0 \Rightarrow s_N \rightarrow \frac{1}{1 - q}$. $\square$

FACT: root test

$\limsup \sqrt[n]{|a_n|} = q < 1$: pick $q < r < 1$; for $n \geq N$, $\sqrt[n]{|a_n|} \leq r \Rightarrow |a_n| \leq r^n \Rightarrow$ comparison with a convergent geometric series. If $q > 1$ then $|a_n| \geq 1$ **infinitely often** $\Rightarrow a_n \not\rightarrow 0$. $\square$

FACT: Cauchy–Hadamard

Apply the root test to $\sum a_n (x - x_0)^n$: $\limsup \sqrt[n]{|a_n| |x - x_0|^n} = |x - x_0| \limsup \sqrt[n]{|a_n|} = \frac{|x - x_0|}{R}$, which is < 1 exactly when $|x - x_0| < R$. $\square$

FACT: comparison test

$0 \leq a_n \leq b_n$ with $\sum b_n < \infty$: the partial sums of $\sum a_n$ are increasing and bounded above by $\sum b_n \Rightarrow$ **monotone + bounded** $\Rightarrow$ convergent. $\square$

FACT: Leibniz criterion

$b_n \downarrow 0$: (s_{2n}) increases, (s_{2n+1}) decreases, $s_{2n} \leq s_{2n+1}$, and $s_{2n+1} - s_{2n} = b_{2n+1} \rightarrow 0 \Rightarrow$ **nested intervals** $\Rightarrow$ both converge to a common s . $\square$

FACT: monotone functions are Riemann integrable

f **increasing** on $[a, b]$, equidistant partition of mesh $\frac{b-a}{n}$: $U(f, P) - L(f, P) = \frac{b-a}{n} \sum_i (f(x_i) - f(x_{i-1})) = \frac{b-a}{n} (f(b) - f(a)) \rightarrow 0$ by **telescoping** $\Rightarrow$ integrable. $\square$

FACT: uniform convergence allows swapping lim and $\int$

$|\int_a^b f_n - \int_a^b f| \leq \int_a^b |f_n - f| \leq (b - a) \|f_n - f\|_\infty \rightarrow 0$. $\square$

10 ODE – 11 Proofs

p7 / 10

FACT: l'Hospital (idea) With $f(x_0) = g(x_0) = 0$, the Cauchy MVT gives $\frac{f(x)}{g(x)} = \frac{f(x)-f(x_0)}{g(x)-g(x_0)} = \frac{f'(\xi_x)}{g'(\xi_x)}$ for some ξ_x between x_0 and x ; $x \rightarrow x_0$ forces $\xi_x \rightarrow x_0$. □
FACT: Archimedean property Suppose not: if $\mathbb{N}$ were bounded above, $s := \sup \mathbb{N}$ would exist; $s - 1$ is not an upper bound $\Rightarrow \exists n > s - 1 \Rightarrow n + 1 > s$, contradicting that s is an upper bound. Hence $\frac{1}{n} \rightarrow 0$. □
FACT: characterisation of the supremum $s = \sup M \iff$ (i) $x \leq s \ \forall x \in M$ and (ii) $\forall \epsilon > 0 \ \exists x \in M : x > s - \epsilon$. “ $\Rightarrow$ ”: if (ii) failed for some ϵ , then $s - \epsilon$ would be a smaller upper bound. “ $\Leftarrow$ ”: (i) makes s an upper bound, (ii) makes it the least one. □
FACT: Cauchy–Schwarz For every $t \in \mathbb{R}$: $0 \leq \ x+ty\ ^2 = \ x\ ^2 + 2t\langle x, y \rangle + t^2\ y\ ^2$. A real quadratic in t that is never negative has discriminant ≤ 0 : $4\langle x, y \rangle^2 - 4\ x\ ^2\ y\ ^2 \leq 0$. Equality forces a double root, i.e. $x + ty = 0$ for some t . □
FACT: convex $\iff f'' \geq 0$ (differentiable case) f convex $\iff$ the chord slopes increase $\iff f'$ is monotonically increasing $\iff f'' \geq 0$ (monotonicity criterion). □
FACT: contraction: uniqueness of the fixed point $ f(x) - f(y) \leq q x - y $ with $q < 1$. Two fixed points give $ x^* - y^* \leq q x^* - y^* \Rightarrow (1 - q) x^* - y^* \leq 0 \Rightarrow x^* = y^*$. Existence: $x_{n+1} = f(x_n)$ satisfies $ x_{n+1} - x_n \leq q^n x_1 - x_0 $, hence is Cauchy. □
FACT: a convergent sequence has exactly one accumulation point $a_n \rightarrow a$: every subsequence has indices $n_k \geq k$, so it also converges to a . Hence a is the only subsequential limit. □

12

Exam block: true/false index and model solutions

12.1 True/false scan index	
TABLE: sequences and series	
Statement	Reason / counterexample
bounded $\Rightarrow$ convergent	F $(-1)^n$
monotone $\Rightarrow$ convergent	F $a_n = n$
monotone + bounded $\Rightarrow$ convergent	T monotone conv. thm
$ a_{n+1} - a_n \rightarrow 0 \Rightarrow$ convergent	F $\sqrt{n}, \sum_{k \leq n} \frac{1}{k}$
Cauchy $\Rightarrow$ convergent (in $\mathbb{R}$)	T completeness
convergent $\Rightarrow$ bounded	T $\epsilon = 1$ argument
monotone $\Rightarrow$ has an accumulation point	F $a_n = n$ has none
monotone $\Rightarrow$ has ≥ 2 accumulation points	F never more than one
bounded $\Rightarrow$ has an accumulation point	T Bolzano–Weierstrass
$\min M$ is an accumulation point	F $x_0 = 0, x_n = 1 + \frac{1}{n}$
$\inf M = \sup M \Rightarrow$ converges	T the sequence is constant
$\sup M$ always exists in $\mathbb{R}$	F unbounded set

TABLE: sequences and series (cont.): series	
Statement	Reason / counterexample
$a_n < b_n \ \forall n \Rightarrow \lim a_n < \lim b_n$	F only $\leq$ survives
$a_n \rightarrow 0 \Rightarrow \sum a_n$ converges	F $\sum \frac{1}{n}$
$\sum a_n$ conv. $\Rightarrow a_n \rightarrow 0$	T divergence test
abs. conv. $\Rightarrow$ conv.	T Cauchy criterion
conv. $\Rightarrow$ abs. conv.	F $\sum \frac{(-1)^n}{n}$
ratio $= 1 \Rightarrow$ diverges	F no information at $q = 1$
$\sum a_n, \sum b_n$ conv. $\Rightarrow \sum a_nb_n$ conv.	F $a_n = b_n = \frac{(-1)^n}{\sqrt{n}}$
reordering never changes the sum	F Riemann, if only conditional

TABLE: functions, continuity, derivatives	
Statement	Reason / counterexample
continuous $\Rightarrow$ differentiable	F $ x $ at 0
differentiable $\Rightarrow$ continuous	T difference quotient
differentiable $\Rightarrow C^1$	F $x^2 \sin \frac{1}{x}$
$f - f(x_0) = O(x - x_0) \Rightarrow$ differentiable	F $ x $ satisfies it
$f - f(x_0) - K(x - x_0) = o(x - x_0) \Rightarrow$ diff.	T that is the definition
one-sided difference quotient exists $\Rightarrow$ diff.	F need both sides
$f'(x_0) = 0 \Rightarrow$ extremum	F x^3 at 0
interior extremum + diff. $\Rightarrow f'(x_0) = 0$	T Fermat
$f''(x_0) = 0 \Rightarrow$ inflection point	F x^4 at 0
continuous on $\mathbb{R} \Rightarrow$ unif. continuous	F $x^2, \sin(x^2)$
continuous on compact $\Rightarrow$ unif. continuous	T Heine
unbounded $f' \Rightarrow$ not unif. continuous	F $\sqrt{x}$ on $[0, 1]$
Lipschitz $\Rightarrow$ unif. continuous	T $\delta = \epsilon/L$
continuous on $(0, 1) \Rightarrow$ attains its max	F $\frac{1}{x}$, not compact

TABLE: functions (cont.): IVT, convexity, polynomials	
Statement	Reason / counterexample
IVT applies on any interval	F needs continuity + compact
convex, interior critical pt $\Rightarrow$ global min	T f' increasing
convex $\Rightarrow$ global min only on the boundary	F it may be interior
convex $\Rightarrow$ global max on the boundary	T chord above graph
convex diff. $\Rightarrow$ several isolated crit. pts	F critical set is an interval
$\deg p = 5 \Rightarrow 5$ distinct roots in $\mathbb{C}$	F multiplicity (z^5)
$\deg p = 5 \Rightarrow 5$ roots with multiplicity	T fund. thm. of algebra
non-real roots come in conjugate pairs	F only for real coefficients
$\deg p$ odd $\Rightarrow$ a real root	F only for real coefficients

TABLE: subsequences, monotonicity, sets – the single-choice favourites	
Statement	Reason / counterexample
every subsequence has a convergent subsequence $\Rightarrow (a_n)$ converges	F true for every bounded seq.: $(-1)^n$
every subsequence converges $\Rightarrow (a_n)$ converges	T (a_n) is a subsequence of itself
monotone + convergent $\Rightarrow$ limit $= \sup M$ (incr.) / $\inf M$ (decr.)	T monotone conv. thm
f cont. monotone unbounded $\Rightarrow$ surjective	F e^x : unbounded only above
f cont. 2π -periodic and monotone $\Rightarrow$ constant	T periodic kills monotonicity
f monotone on $[a, b] \Rightarrow$ max and min at the endpoints	T no continuity needed
f incr., $f(-1) = -1, f(1) = 1 \Rightarrow \exists x_0 : f(x_0) = 0$	F IVT needs continuity; a jump skips 0
f convex and concave $\Rightarrow$ constant	F affine: $f'' \equiv 0$, not $f' \equiv 0$
$A = \{g > c\}, g$ cont. $\Rightarrow A$ closed	F strict inequality $\Rightarrow$ open

TABLE: integration and function sequences	
Statement	Reason / counterexample
differentiable on $[a, b] \Rightarrow$ integrable	T diff. $\Rightarrow$ cont. $\Rightarrow$ integrable
integrable $\Rightarrow$ no discontinuities	F jumps are fine
integrable $\Rightarrow$ differentiable	F step function
finitely many values $\Rightarrow$ integrable	F Dirichlet 1_q
finitely many discontinuities + bdd $\Rightarrow$ integrable	T Lebesgue criterion
monotone on $[a, b] \Rightarrow$ integrable	T telescoping $U - L$
continuous on $[a, b] \Rightarrow$ integrable	T uniform continuity
unbounded f can be Riemann integrable	F that is improper

TABLE: function sequences and improper integrals	
Statement	Reason / counterexample
pointwise $\Rightarrow$ uniform	F x^n on $[0, 1]$
uniform $\Rightarrow$ pointwise	T same N works
pointwise, all f_n cont. $\Rightarrow f$ cont.	F x^n on $[0, 1]$
uniform, all f_n cont. $\Rightarrow f$ cont.	T $\frac{\epsilon}{5}$ -argument
uniform, f_n cont. $\Rightarrow f$ integrable	T f is continuous
$f_n \rightrightarrows f \Rightarrow f'_n \rightarrow f'$	F $\frac{\sin(nx)}{n}$
pointwise $\Rightarrow \int f_n \rightarrow \int f$	F n^2xe^{-nx} on $[0, 1]$
$\sup f_n \geq \frac{1}{2n} \Rightarrow$ cannot conv. unif. to 0	F $\frac{1}{2n} \rightarrow 0$ anyway
max at a different x_n each $n \Rightarrow$ not uniform	F only the value matters
$\int_0^\infty f$ conv. $\Rightarrow$ check only ∞	F $\frac{\sin t}{t}$ blows up at 0
$f > 0, \int_1^\infty f < \infty \Rightarrow f \rightarrow 0$	F thin spikes of height 1
$\int_1^\infty \frac{\cos^2 t}{t} dt$ converges	F $\cos^2 t = \frac{1+\cos 2t}{2}$ hides $\int \frac{dt}{2t}$
$\int_1^\infty \frac{\cos t}{t} dt$ converges	T conditionally, by Dirichlet
$\int_{-1}^1 \frac{dx}{x} = 0$ by symmetry	F both halves must converge separately
$y' = f(x, y), f$ continuous $\Rightarrow$ unique solution	F $y' = \sqrt{ y }, y(0) = 0$

TABLE: power series, Taylor, ODE	
Statement	Reason / counterexample
R is determined by a_n alone	T Cauchy–Hadamard
convergence on $ x - x_0 < R$ is absolute	T root test with $q < 1$
convergence is uniform on $ x - x_0 < R$	F only on closed $[x_0 - r, x_0 + r]$, $r < R$
the two endpoints behave the same	F $\sum \frac{x^n}{n}$: -1 conv., 1 div.
inside R you may differentiate termwise	T and R is unchanged
$C^\infty \Rightarrow$ equal to its Taylor series	F needs $R_n \rightarrow 0$
Taylor coefficients are unique	T $a_n = \frac{f^{(n)}(x_0)}{n!}$
T_n error $\leq \frac{\max f^{(n+1)} }{(n+1)!} x - x_0 ^{n+1}$	T Lagrange remainder
order n ODE $\Rightarrow n$ free constants	T fundamental system

TABLE: ODE: existence, uniqueness, ansatz	
Statement	Reason / counterexample
IVP for a linear ODE always uniquely solvable	T continuous coefficients
BVP always uniquely solvable	F $0, 1$ or ∞ many
μ = root of char. poly. $\Rightarrow$ ansatz needs x^m	T resonance
$g = \cos \omega x \Rightarrow$ ansatz $A \cos \omega x$ suffices	F need $A \cos + B \sin$
separable ODE: constant solutions can be skipped	F they are solutions too
$y' = f(x, y)$, f Lipschitz in $y \Rightarrow$ unique	T Picard–Lindelöf
autonomous: solutions can cross an equilibrium	F uniqueness forbids it

12.2 Exam-day checklist

TRAP: mistakes that cost marks in written proofs	
- Using a theorem without checking its hypotheses — above all <i>compactness</i> for IVT/<i>Weierstrass/Heine</i>, and <i>monotonicity</i> for <i>Leibniz</i>. - Applying l'Hospital without first verifying the form is $\frac{0}{0}$ or $\frac{\infty}{\infty}$. - Writing lim of a product/quotient as the product/quotient of limits when one of them does not exist. - Choosing N or δ that secretly depends on the wrong variable — in a <i>uniform</i> statement, δ may not depend on the point. - Proving one direction of an “$\Longleftrightarrow$” and stopping. - Concluding “$<$” from a limit: $a_n < b_n$ only gives $\lim a_n \leq \lim b_n$. - Dividing by something that can vanish (e.g. $g(y)$ in a separable ODE) without treating that case. - Forgetting the constant solutions, the $+C$, or the second endpoint of an improper integral. - Asserting a sup/max exists without invoking completeness or <i>Weierstrass</i>. - Doing algebra instead of stating the argument: name the theorem, that is where the marks are.	
TRAP: read this in the first minute	
- Integral asked, options given? Differentiate the options instead of integrating (§9.1). - Limit as $x \rightarrow 0$? Taylor beats l'Hospital almost always (§5.3 vs §8.5). - Improper integral? Split first, check <i>both</i> endpoints (§9.8). - Separable ODE? Write down the constant solutions $g(y) = 0$ (§10.2). - Forced linear ODE? Resonance check before the ansatz (§10.6). “Zeigen Sie, dass es konvergiert”? Name the theorem (monotone + bounded / Cauchy / comparison, §4.2). Naming earns marks. “Begründen Sie”? One sentence per step, and quote the hypotheses of every theorem you cite. - Substitution finished? Differentiate the result; the chain-rule factor must reappear (§9.2). - Uniform convergence? Compute sup $f_n - f$ explicitly (§7.2); a maximiser x_n is already a complete answer for “no”. - Stuck on a proof? Write out the definition of everything given and everything to be shown. Often half the marks. - Multiple choice, no negative marking: never leave a box empty.	

12.3 Model solutions for the open questions

EXAMPLE: “Untersuchen Sie $f_n(x) = \frac{x}{1+nx^2}$ auf $\mathbb{R}$ ” (9 P)	
a) Gleichmässige Stetigkeit. For $n = 0$, $f_0(x) = x$ is Lipschitz with constant 1, hence uniformly continuous. For $n \geq 1$, f_n is differentiable on $\mathbb{R}$ with $f'_n(x) = \frac{1-nx^2}{(1+nx^2)^2}$, so $f'_n(x) \leq \frac{1+nx^2}{(1+nx^2)^2} = \frac{1}{1+nx^2} \leq 1$. By the mean value theorem $f_n(x) - f_n(y) \leq x - y$ for all x, y, so $\delta := \varepsilon$ satisfies the definition. Hence every f_n is uniformly continuous. b) Punktwiser Grenzwert. Fix x. For $x = 0$, $f_n(0) = 0$ for all n. For $x \neq 0$, $f_n(x) = \frac{ x }{1+nx^2} \leq \frac{ x }{nx^2} = \frac{1}{n x } \rightarrow 0$. Hence $f_n \rightarrow f \equiv 0$ pointwise on $\mathbb{R}$. c) Gleichmässige Konvergenz. $\ f_n - f\ _\infty = \sup_x f_n(x)$. From $f'_n(x) = 0$ we get $x = \pm \frac{1}{\sqrt{n}}$, with $f_n(\frac{1}{\sqrt{n}}) = \frac{1}{2\sqrt{n}}$, and $f_n(x) \rightarrow 0$ as $x \rightarrow \infty$. Therefore $\ f_n\ _\infty = \frac{1}{2\sqrt{n}} \rightarrow 0$, i.e. $f_n \rightrightarrows 0$ uniformly on all of $\mathbb{R}$. $\square$	
EXAMPLE: “ $f(x) = \frac{\sin x}{1+x^2}$: Fixpunkt und Rekursion” (9 P)	
a) Unique fixed point, and $f(x) < x$ for $x \neq 0$. $f(0) = 0$, so 0 is a fixed point. For $x \neq 0$ use the strict inequality $\sin x < x$ together with $1 + x^2 \geq 1$: $ f(x) = \frac{ \sin x }{1+x^2} \leq \sin x < x .$ If $x^* \neq 0$ were another fixed point then $x^* = f(x^*) < x^*$, a contradiction. So $x^* = 0$ is the unique fixed point. b) Behaviour of $x_{n+1} = f(x_n)$, $x_0 \in (0, 1)$. <i>Positivity</i>: for $x \in (0, 1)$ we have $\sin x > 0$, so $f(x) \in (0, x) \subseteq (0, 1)$; by induction $x_n \in (0, 1)$ for all n. <i>Monotonicity</i>: by part a), $x_{n+1} = f(x_n) < x_n$, so (x_n) is strictly decreasing. <i>Conclusion</i>: decreasing and bounded below by 0, hence convergent by the monotone convergence theorem; let $L := \lim x_n \geq 0$. Since f is continuous, $L = \lim x_{n+1} = \lim f(x_n) = f(L)$, so L is a fixed point, and by a) $L = 0$. Thus $x_n \rightarrow 0$ for every $x_0 \in (0, 1)$. $\square$	
EXAMPLE: box answers (3 P each, only the result is marked)	
Limit. $\lim_{x \rightarrow 0} \frac{\sin x - x + x^3/6}{x^5}$: insert $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \dots \Rightarrow$ numerator $= \frac{x^5}{120} + O(x^7) \Rightarrow \frac{1}{120}$. ODE. $y'' - y = e^x$: $\lambda = \pm 1$; $\mu = 1$ is a simple root $\Rightarrow$ resonance $\Rightarrow y_p = A x e^x$, $y_p'' - y_p = 2A e^x \Rightarrow A = \frac{1}{2} \Rightarrow y = C_1 e^x + C_2 e^{-x} + \frac{1}{2} x e^x$. Integral. $\int \frac{dx}{1+\cos x}$: $1 + \cos x = 2 \cos^2 \frac{x}{2}$, then $u = \frac{x}{2}$, $dx = 2 du \Rightarrow \int \frac{2 du}{\cos^2 u} = \tan u \Rightarrow \tan \frac{x}{2} + C$. Radius. $\sum \frac{n^2 (x-2)^n}{(2n)!}$: coefficient ratio $\rightarrow 0 \Rightarrow R = \infty \Rightarrow$ converges for all $x \in \mathbb{R}$. Improper: $\int_1^\infty \frac{\cos t}{t} dt$ conv.; $\int_1^\infty \frac{\cos^2 t}{t} dt$ div.; $\int_0^\infty \frac{\sin t}{t^3} dt$ div. (at $t = 0$).	
EXAMPLE: “Für welche x konvergiert die Reihe?” (radius + endpoints)	
For $\sum_{n \geq 1} \frac{(x-3)^n}{n 2^n}$: the centre is $x_0 = 3$ and $a_n = \frac{1}{n 2^n}$. Radius: $\left \frac{a_{n+1}}{a_n} \right = \frac{n 2^n}{(n+1) 2^{n+1}} = \frac{n}{2(n+1)} \rightarrow \frac{1}{2}$, hence $R = 2$. By Cauchy–Hadamard the series converges absolutely for $x - 3 < 2$ and diverges for $x - 3 > 2$. Endpoint $x = 5$: the series becomes $\sum \frac{2^n}{n 2^n} = \sum \frac{1}{n}$, the harmonic series, which diverges. Endpoint $x = 1$: it becomes $\sum \frac{(-2)^n}{n 2^n} = \sum \frac{(-1)^n}{n}$, which converges by the Leibniz criterion ($\frac{1}{n}$ decreases monotonically to 0), though not absolutely. Answer: the series converges exactly for $x \in [1, 5)$. $\square$! Always test the two endpoints <i>separately</i>; they need not agree.	
EXAMPLE: “Konvergiert das uneigentliche Integral?” (write-up)	
For $\int_0^\infty \frac{dx}{\sqrt{x}(1+x)}$ there are two problem points, so we split: $\int_0^\infty = \int_0^1 + \int_1^\infty$, and both pieces must converge. At $x \rightarrow 0^+$: $\frac{1}{\sqrt{x}(1+x)} \sim \frac{1}{\sqrt{x}} = x^{-1/2}$, and $\int_0^1 x^{-p} dx$ converges for $p < 1$; here $p = \frac{1}{2} < 1$, so $\int_0^1$ converges. At $x \rightarrow \infty$: $\frac{1}{\sqrt{x}(1+x)} \sim x^{-3/2}$, and $\int_1^\infty x^{-p} dx$ converges for $p > 1$; here $p = \frac{3}{2} > 1$, so $\int_1^\infty$ converges. Both pieces converge, hence the integral converges. (Its value is π, via $x = t^2$.) $\square$! The exponent threshold <i>reverses</i> between the two ends: mild singularity allowed at 0, fast decay required at ∞.	
EXAMPLE: “Bestimmen Sie Extrema und Wendepunkte” (curve discussion write-up)	
For $f(x) = \frac{x}{1+x^2}$: Domain $\mathbb{R}$, f is odd and continuous as a quotient of continuous functions with denominator $\neq 0$. First derivative $f'(x) = \frac{1-x^2}{(1+x^2)^2}$, so $f'(x) = 0 \iff x = \pm 1$. Second derivative $f''(x) = \frac{2x(x^2-3)}{(1+x^2)^3}$; $f''(1) = -\frac{1}{2} < 0$, hence $x = 1$ is a local maximum with $f(1) = \frac{1}{2}$, and $f''(-1) = \frac{1}{2} > 0$, hence $x = -1$ is a local minimum with $f(-1) = -\frac{1}{2}$. <i>Monotonicity</i> $f' > 0$ on $(-1, 1)$ and $f' < 0$ outside, so f increases on $[-1, 1]$ and decreases on $(-\infty, -1]$ and $[1, \infty)$. <i>Inflection points</i> f'' changes sign at $x = 0, \pm\sqrt{3}$. <i>Asymptote</i> $\lim_{x \rightarrow \pm\infty} f(x) = 0$, so $y = 0$. Since $f \rightarrow 0$ at $\pm\infty$ and the only candidates are ± 1, the local extrema are also <i>global</i>: $\max f = \frac{1}{2}$, $\min f = -\frac{1}{2}$, both attained. $\square$	

13

Lookup tables

13.1

How to use the derivative / antiderivative table

TRAP: read this before you copy a single row

Every row means $\frac{d}{dx}F(x) = f(x)$ and $\int f(x) dx = F(x) + C$, written for the **bare argument** x .

1. Inner function linear, $ax + b$: $\int f(ax + b) dx = \frac{1}{a}F(ax + b) + C$ — **divide by the inner derivative** a .

2. Inner function anything else: you may **not** patch with a factor. **Substitute** $u = g(x)$, $dx = \frac{du}{g'(x)}$, and continue only if the leftover $g'(x)$ cancels.

3. Constant prefactors stay outside: $\int cf = c \int f$. Do the prefactor and the substitution factor as *separate written steps*; never patch by feeling.

4. **Differentiate your answer. The chain-rule factor must reappear.**

The trap in full: $\int \frac{dx}{2 \cos^2(x/2)}$. With $u = \frac{x}{2}$, $dx = 2 du$: $\int \frac{2 du}{2 \cos^2 u} = \tan u = \tan \frac{x}{2} + C$ — the 2 from dx cancels the $\frac{1}{2}$ in front, so the answer is **not** $2 \tan \frac{x}{2}$.

13.2

Derivatives and antiderivatives

TABLE: powers, roots, exponentials, logarithms		
$F = \int f$	$f(x)$	$f'(x)$
$\frac{x^{n+1}}{n+1}$ ($n \neq -1$)	x^n	nx^{n-1}
$\ln x $	$\frac{1}{x}$	$-\frac{1}{x^2}$
$\frac{2}{3} x^{3/2}$	$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$
$\frac{n}{n+1} x^{(n+1)/n}$	$\sqrt[n]{x}$	$\frac{1}{n} x^{1/n-1}$
$\frac{(ax+b)^{n+1}}{a(n+1)}$	$(ax+b)^n$	$an(ax+b)^{n-1}$
e^x	e^x	e^x
$\frac{1}{a} e^{ax}$	e^{ax}	$a e^{ax}$
$\frac{a^x}{\ln a}$	a^x	$a^x \ln a$
$x(\ln x - 1)$	$\ln x $	$\frac{1}{x}$
$\frac{x}{a} (\ln x - 1)$	$\log_a x $	$\frac{1}{x \ln a}$

TABLE: trigonometric and inverse trigonometric		
$F = \int f$	$f(x)$	$f'(x)$
$-\cos x$	$\sin x$	$\cos x$
$\sin x$	$\cos x$	$-\sin x$
$-\ln \cos x $	$\tan x$	$\frac{1}{\cos^2 x} = 1 + \tan^2 x$
$\ln \sin x $	$\cot x$	$-\frac{1}{\sin^2 x}$
$\ln \left \tan \frac{x}{2} \right $	$\frac{1}{\sin x}$	$-\frac{\cos x}{\sin^2 x}$
$\ln \left \tan \left(\frac{x}{2} + \frac{\pi}{4} \right) \right $	$\frac{1}{\cos x}$	$\frac{\sin x}{\cos^2 x}$
$\frac{x}{2} - \frac{\sin 2x}{4}$	$\sin^2 x$	$\sin 2x$
$\frac{x}{2} + \frac{\sin 2x}{4}$	$\cos^2 x$	$-\sin 2x$
$x \arcsin x + \sqrt{1-x^2}$	$\arcsin x$	$\frac{1}{\sqrt{1-x^2}}$
$x \arccos x - \sqrt{1-x^2}$	$\arccos x$	$-\frac{1}{\sqrt{1-x^2}}$
$x \arctan x - \frac{1}{2} \ln(1+x^2)$	$\arctan x$	$\frac{1}{1+x^2}$

TABLE: hyperbolic		
$F = \int f$	$f(x)$	$f'(x)$
$\cosh x$	$\sinh x$	$\cosh x$
$\sinh x$	$\cosh x$	$\sinh x$
$\ln \cosh x$	$\tanh x$	$\frac{1}{\cosh^2 x}$
$x \operatorname{arsinh} x - \sqrt{1+x^2}$	$\operatorname{arsinh} x$	$\frac{1}{\sqrt{1+x^2}}$
$x \operatorname{arcosh} x - \sqrt{x^2-1}$	$\operatorname{arcosh} x$	$\frac{\sqrt{x^2-1}}{x^2}$
$x \operatorname{artanh} x + \frac{1}{2} \ln(1-x^2)$	$\operatorname{artanh} x$	$\frac{1}{1-x^2}$
$\sinh x = \frac{e^x - e^{-x}}{2}$, $\cosh x = \frac{e^x + e^{-x}}{2}$, $\cosh^2 - \sinh^2 = 1$;		
$\operatorname{arsinh} x = \ln(x + \sqrt{x^2+1})$, $\operatorname{arcosh} x = \ln(x + \sqrt{x^2-1})$, $\operatorname{artanh} x = \frac{1}{2} \ln \frac{1+x}{1-x}$.		

TABLE: antiderivatives you will not guess (1)	
$f(x)$	$F(x) = \int f\,dx$
$\frac{1}{f(x)}$	$\frac{1}{a} \ln ax + b $
$\frac{1}{a^2+x^2}$	$\ln f(x) $
$\frac{1}{a^2-x^2}$	$\frac{1}{a} \arctan \frac{x}{a}$
$\frac{1}{\sqrt{a^2-x^2}}$	$\frac{1}{2a} \ln \left \frac{a+x}{a-x} \right $
$\frac{1}{\sqrt{x^2-a^2}}$	$\arcsin \frac{x}{a}$
$\ln x$	$\ln \left x + \sqrt{x^2 \pm a^2} \right $
$\frac{1}{1+\cos x}$	$x \ln x - x$
$\frac{1}{1-\cos x}$	$\tan \frac{x}{2}$
$\frac{1}{\sin x \cos x}$	$-\cot \frac{x}{2}$
	$\ln \tan x $

TABLE: antiderivatives you will not guess (2)	
$f(x)$	$F(x) = \int f\,dx$
$\sqrt{a^2-x^2}$	$\frac{x}{2} \sqrt{a^2-x^2} + \frac{a^2}{2} \arcsin \frac{x}{a}$
$\sqrt{a^2+x^2}$	$\frac{x}{2} \sqrt{a^2+x^2} + \frac{a^2}{2} \ln \left x + \sqrt{a^2+x^2} \right $
$\sqrt{x^2-a^2}$	$\frac{x}{2} \sqrt{x^2-a^2} - \frac{a^2}{2} \ln \left x + \sqrt{x^2-a^2} \right $
xe^{ax}	$\frac{e^{ax}}{a^2} (ax - 1)$
$e^{ax} \sin bx$	$\frac{e^{ax}}{a^2+b^2} (a \sin bx - b \cos bx)$
$e^{ax} \cos bx$	$\frac{e^{ax}}{a^2+b^2} (a \cos bx + b \sin bx)$
$x \sin x$	$\sin x - x \cos x$
$x \cos x$	$\cos x + x \sin x$
x^x	no elementary form; $\frac{d}{dx} x^x = x^x (1 + \ln x)$

13.3 Limits, growth, series

FACT: growth hierarchy — kills most limits instantly
As $x \rightarrow \infty$, for $\alpha > 0$ and $b > 1$:
$\ln x \ll x^\alpha \ll b^x \ll x! \ll x^x$
Every ratio of a term on the left to one on the right tends to 0.

TABLE: standard limits	
$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$	$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$
$\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$	$\lim_{x \rightarrow 0} \frac{\arctan x}{x} = 1$
$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$	$\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln a$
$\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$	$\lim_{x \rightarrow 1} \frac{\ln x}{x-1} = 1$
$\lim_{x \rightarrow 0^+} x \ln x = 0$	$\lim_{x \rightarrow 0^+} x^x = 1$
$\lim_{x \rightarrow \infty} \frac{\ln x}{x^\alpha} = 0$	$\lim_{x \rightarrow \infty} \frac{x^m}{e^x} = 0$
$\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$	$\lim_{n \rightarrow \infty} \sqrt[n]{a} = 1 \ (a > 0)$
$\lim_{n \rightarrow \infty} \frac{a^n}{n!} = 0$	$\lim_{n \rightarrow \infty} \sqrt[n]{n!} = \infty$
$\lim_{x \rightarrow 0} (1+x)^{1/x} = e$	$\lim_{x \rightarrow \pm \infty} \left(1 + \frac{k}{x}\right)^{mx} = e^{km}$
$\lim_{x \rightarrow \infty} x^\alpha q^x = 0 \ (q < 1)$	$\lim_{x \rightarrow \infty} \arctan x = \frac{\pi}{2}$

TABLE: Taylor series at 0, with radius	
$e^x = \sum_{n \geq 0} \frac{x^n}{n!}, R = \infty$ $\sin x = \sum_{n \geq 0} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{6} + \frac{x^5}{120} - \dots, R = \infty$ $\cos x = \sum_{n \geq 0} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \dots, R = \infty$ $\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots, R = \frac{\pi}{2}$ $\frac{1}{1-x} = \sum_{n \geq 0} x^n, R = 1$ $\ln(1+x) = \sum_{n \geq 1} \frac{(-1)^{n+1} x^n}{n}, R = 1$ $\arctan x = \sum_{n \geq 0} \frac{(-1)^n x^{2n+1}}{2n+1}, R = 1$ $(1+x)^\alpha = \sum_{n \geq 0} \binom{\alpha}{n} x^n, R = 1$ $\sinh x = \sum \frac{x^{2n+1}}{(2n+1)!}, \cosh x = \sum \frac{x^{2n}}{(2n)!}, R = \infty$	

13.4 Trigonometry, logs, algebra

TABLE: trigonometric identities	
$\sin^2 x + \cos^2 x = 1; 1 + \tan^2 x = \frac{1}{\cos^2 x}$ $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$ $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$ $\sin 2x = 2 \sin x \cos x; \cos 2x = \cos^2 x - \sin^2 x = 1 - 2 \sin^2 x$ $\sin^2 x = \frac{1 - \cos 2x}{2}; \cos^2 x = \frac{1 + \cos 2x}{2}$ $1 + \cos x = 2 \cos^2 \frac{x}{2}; 1 - \cos x = 2 \sin^2 \frac{x}{2}$ $\tan \frac{x}{2} = \frac{\sin x}{1 + \cos x} = \frac{1 - \cos x}{\sin x}$ $\sin(-x) = -\sin x$ (odd); $\cos(-x) = \cos x$ (even) $\sin(\pi - x) = \sin x, \cos(\pi - x) = -\cos x$ $\sin\left(x + \frac{\pi}{2}\right) = \cos x, \cos\left(x + \frac{\pi}{2}\right) = -\sin x$ $\arcsin: [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \arccos: [-1, 1] \rightarrow [0, \pi], \arctan: \mathbb{R} \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$	
Forgot one? Derive it from e^{ix} (§2.1) in ten seconds.	

TABLE: special values and quadrant signs				
deg	rad	sin	cos	tan
0	0	0	1	0
30	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
45	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
90	$\frac{\pi}{2}$	1	0	—
120	$\frac{2\pi}{3}$	$\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$-\sqrt{3}$
135	$\frac{3\pi}{4}$	$\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$	-1
150	$\frac{5\pi}{6}$	$\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{3}}{3}$
180	π	0	-1	0

Quad.	sin	cos	tan
I	+	+	+
II	+	-	-
III	-	-	+
IV	-	+	-

Solving $\sin x = c$: $x = \arcsin c + 2k\pi$ and $x = \pi - \arcsin c + 2k\pi$.
 $\cos x = c$: $x = \pm \arccos c + 2k\pi$. $\tan x = c$: $x = \arctan c + k\pi$.

TABLE: logs, powers, algebra survival kit	
$\log_a x = \frac{\ln x}{\ln a}; a^x = e^{x \ln a}; u^v = e^{v \ln u}$ $\ln(xy) = \ln x + \ln y; \ln \frac{x}{y} = \ln x - \ln y; \ln(x^y) = y \ln x$ $\ln 1 = 0, \ln e = 1; \ln x$ defined only for $x > 0$ $\sqrt{A} - \sqrt{B} = \frac{A-B}{\sqrt{A}+\sqrt{B}}$ (conjugate trick, kills $\infty - \infty$) $a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + \dots + b^{n-1})$ - complete the square: $x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + c - \frac{b^2}{4}$ $\binom{n}{k} = \frac{n!}{k!(n-k)!}; (a+b)^n = \sum_k \binom{n}{k} a^k b^{n-k}$ $\sum_{k=1}^n k = \frac{n(n+1)}{2}; \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}; \sum_{k=0}^n q^k = \frac{1-q^{n+1}}{1-q}$ - guess a polynomial root among the divisors of the constant term, then divide	

13.5 Decision tables and glossary

TABLE: I need to... → use	
show a sequence converges without knowing the limit	monotone + bounded, or Cauchy
show a value is attained	IVT, or Weierstrass max/min
show a zero or fixed point exists	IVT applied to $f(x) - x$
show it is unique	strict monotonicity, or Rolle, or contraction
classify a critical point	sign of f'' ; if $f'' = 0$, sign change of f'
evaluate $\frac{0}{0}$ or $\frac{\infty}{\infty}$	Taylor (preferred) or l'Hospital
handle $0 \cdot \infty, \infty \cdot \infty, \infty - \infty, 1^\infty$	rewrite first, then l'Hospital
decide $\int_a^\infty f$ converges	compare with x^{-p} , check both ends
integrate $\sqrt{a^2 \pm x^2}$	trigonometric substitution
integrate a rational function	partial fractions
integrate rational in sin, cos	half-angle, else $t = \tan \frac{x}{2}$
solve a forced linear ODE	$y_h + y_p$, resonance check
solve a 1st order nonlinear ODE	separation / $u = y/x$ / Bernoulli / int. factor
show $f_n \rightrightarrows f$	$\sup f_n - f \rightarrow 0$
disprove $f_n \rightrightarrows f$	one x_n with $ f_n(x_n) - f(x_n) \geq \varepsilon_0$
swap lim and $\int$	needs uniform convergence
prove an inequality	MVT, convexity/tangent line, or induction

TABLE: series looks like... → take	
$a_n \not\rightarrow 0$	divergence test, done
$\sum \frac{1}{n^p}, \sum q^n$	p -series / geometric
factorials	ratio test
n -th powers	root test
alternating with $b_n \downarrow 0$	Leibniz
resembles a known series	comparison on leading powers
$a_n = f(n)$ with $f \downarrow$	integral test
$a_n b_n, b_n \downarrow 0, \sum a_n$ bounded	Abel/Dirichlet
power series	Cauchy–Hadamard + endpoints

TABLE: “für welche α ?” — every parameter threshold in one place	
Object	Behaves well exactly when
$\sum_{n \geq 1} \frac{1}{n^\alpha}$	converges $\iff \alpha > 1$
$\sum_{n \geq 1} \frac{(-1)^n}{n^\alpha}$	$\alpha > 1$ absolutely ; $0 < \alpha \leq 1$ conditionally (Leibniz); $\alpha = 0$ diverges ($a_n \not\rightarrow 0$)
$\sum_{n \geq 2} \frac{1}{n(\log n)^\alpha}$	converges $\iff \alpha > 1$ (integral test)
$\frac{(\log n)^\alpha}{n^\beta}$	$\rightarrow 0 \iff \beta > 0$, for every α — any power of n beats any power of log
$\int_0^1 x^{-\alpha} dx$	converges $\iff \alpha < 1$ (mild singularity at 0)
$\int_1^\infty x^{-\alpha} dx$	converges $\iff \alpha > 1$ (fast decay at ∞)
$x^\alpha \sin \frac{1}{x}$ at $x = 0$	$\rightarrow 0$ and continuous $\iff \alpha > 0$; differentiable $\iff \alpha > 1; C^1 \iff \alpha > 2$
$ x ^\alpha$ at $x = 0$	differentiable $\iff \alpha > 1$ (or $\alpha = 0$)

! The two integral rows have **opposite** thresholds — that is the single most common sign error. A $\sum$ or $\int$ with *both* a singularity and an infinite end can therefore never converge for a pure power.

TABLE: word list DE → EN	
Folge = sequence • Reihe = series • Grenzwert = limit • Häufungspunkt = accumulation point • beschränkt = bounded • monoton = monotone • Teilfolge = subsequence • Schranke = bound • stetig = continuous • gleichmäßig = uniform (ly) • punktweise = pointwise • differenzierbar = differentiable • Ableitung = derivative • Stammfunktion = antiderivative • Zwischenwertsatz = IVT • Mittelwertsatz = MVT • Umkehrfunktion = inverse • Nullstelle = root, zero • Wendepunkt = inflection point • Extremstelle = extremum • Abschätzung = estimate • oBdA = w.l.o.g. • Voraussetzung = hypothesis • Behauptung = claim • Widerspruch = contradiction • Anfangswertproblem = IVP • Randwertproblem = BVP • Trennung der Variablen = separation of variables • uneigentlich = improper • Konvergenzradius = radius of convergence • Partialsumme = partial sum • Partialbruchzerlegung = partial fractions • Kettenregel = chain rule • Produktregel = product rule • konvex = convex • Treppenfunktion = step function • Zerlegung = partition • Obersumme/Untersumme = upper/lower sum.	

FACT: the implication map – which arrow reverses, and which never does

Read $\Rightarrow$ as “always”; a crossed arrow means there is a standard counterexample.

- **Sequences:** convergent $\Rightarrow$ Cauchy $\Rightarrow$ bounded $\Rightarrow$ has an accumulation point. **! Only Cauchy $\Leftrightarrow$ convergent** reverses (completeness, §3.5).
- **Series:** $\sum |a_n| < \infty \Rightarrow \sum a_n$ converges $\Rightarrow a_n \rightarrow 0$. **! Neither reverses:** $\sum \frac{(-1)^n}{n}$, resp. $\sum \frac{1}{n}$.
- **Functions on an interval:** $C^1 \Rightarrow$ differentiable $\Rightarrow$ continuous $\Rightarrow$ Riemann integrable on $[a, b]$. **! None reverses:** $x^2 \sin \frac{1}{x}$, $|x|$, a step function.
- **Continuity strength:** Lipschitz $\Rightarrow$ uniformly continuous $\Rightarrow$ continuous. **! Reverses only on a compact domain, and only the second arrow** (Heine, §6.4).
- **Function sequences:** $f_n \rightrightarrows f \Rightarrow f_n \rightarrow f$ pointwise, and $\Rightarrow \int f_n \rightarrow \int f$ on $[a, b]$. **! Uniform convergence says **nothing** about f'_n** (§7.3).
- **Monotone + bounded** is the one hypothesis pair that upgrades “bounded” all the way to “convergent”, and it is the workhorse for every recursive sequence.

TABLE: writing it in German — sentences that earn the marks	
You want to say	Write
start an ε proof	<i>Sei $\varepsilon > 0$ beliebig.</i>
choose the index	<i>Wähle $N := \lceil 1/\varepsilon \rceil$. Dann gilt für alle $n \geq N$:</i>
cite a hypothesis	<i>Nach Voraussetzung ist f auf $[a, b]$ stetig.</i>
cite a theorem	<i>Nach dem Zwischenwertsatz existiert ein $x^* \in [a, b]$ mit ...</i>
w.l.o.g.	<i>ObdA sei $a < b$ (sonst vertausche die Rollen).</i>
contradiction	<i>Angenommen, ... Widerspruch zu ... Also gilt ...</i>
induction	<i>Induktionsanfang / Induktionsvoraussetzung / Induktionsschritt.</i>
conclude	<i>Daraus folgt / Somit ist gezeigt, dass ... □</i>
Marks come from naming things: every theorem you use by name, with its hypotheses quoted, is worth a point on its own (§12.2).	